

BFS Traversal

Step 1: Start at A (Seeking target node K)

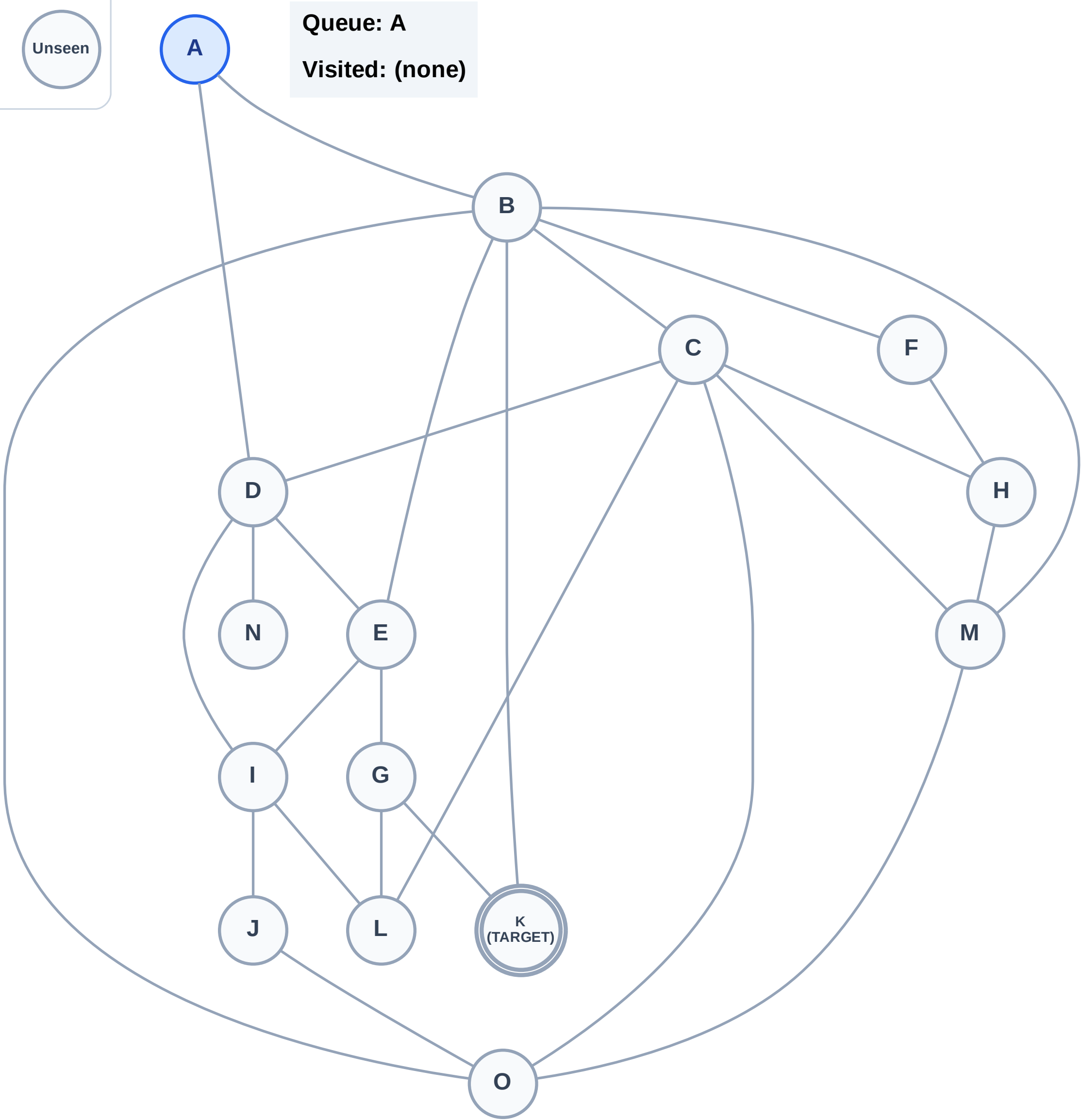
Legend

Complete

Processing

Discovered

Unseen



BFS Traversal

Step 2: Process node A

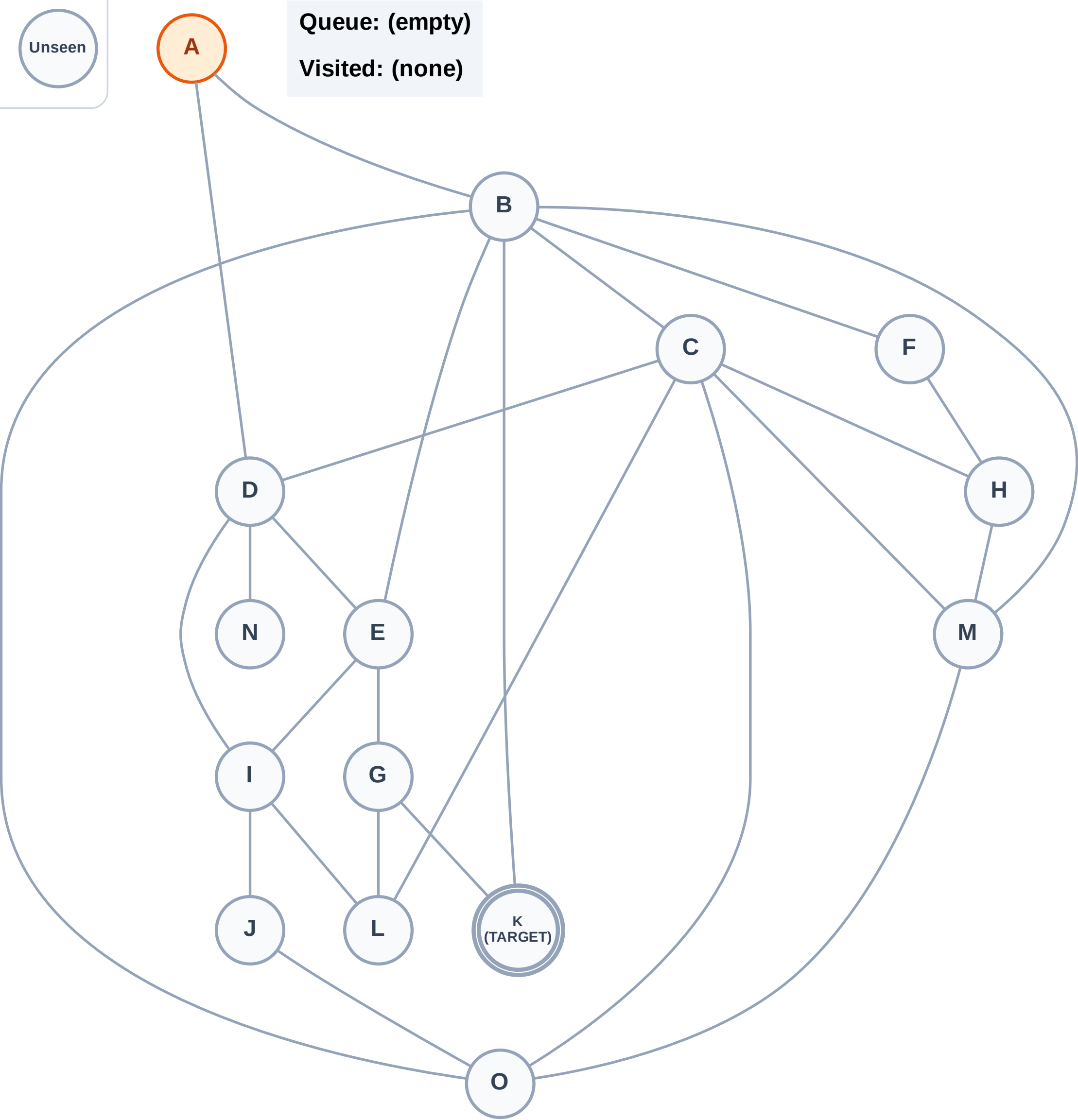
Legend

Complete

Processing

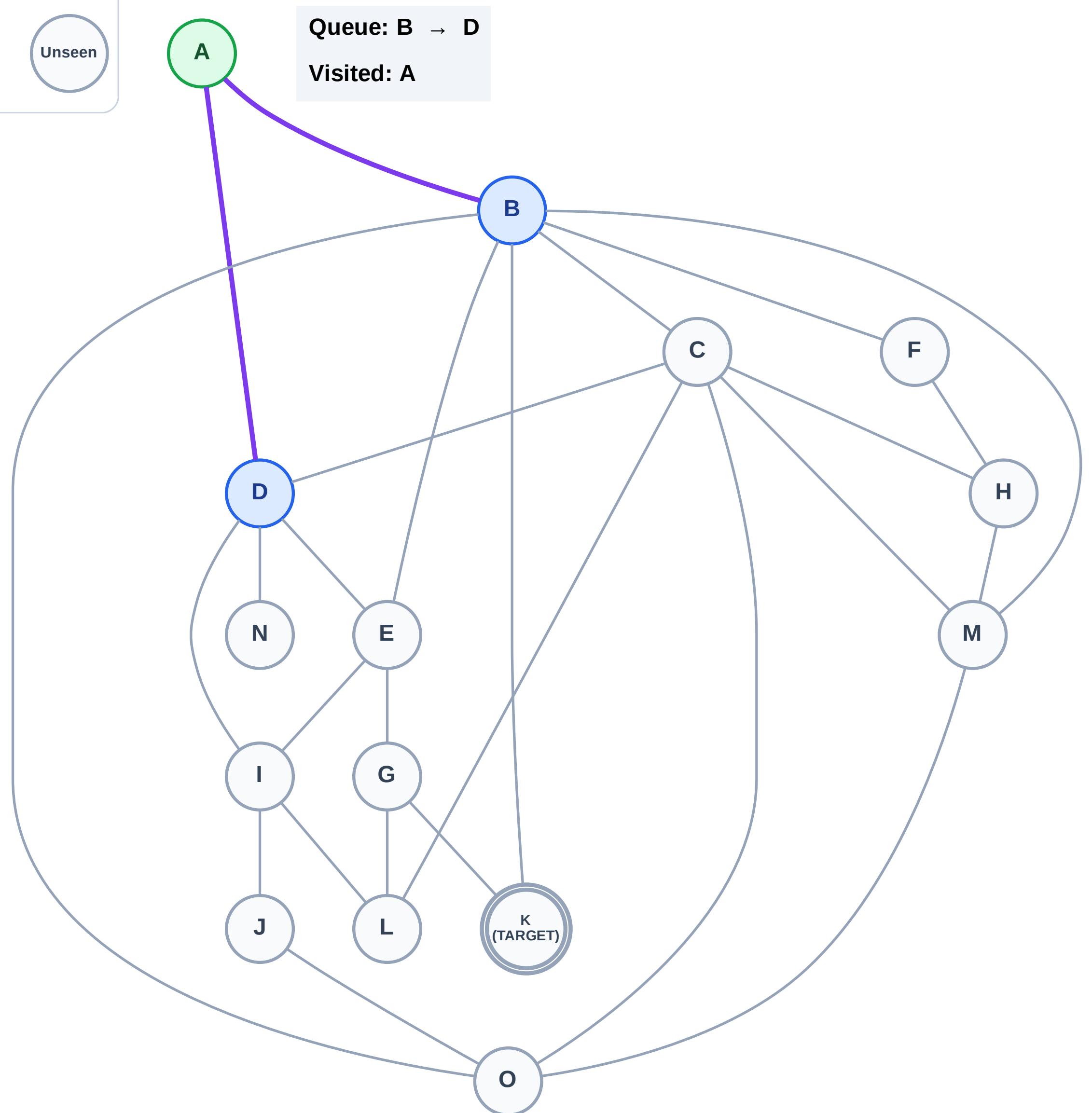
Discovered

Unseen



Step 3: Discover B, D from A

Step 3: Discover B, D from A



BFS Traversal

Step 4: Process node B

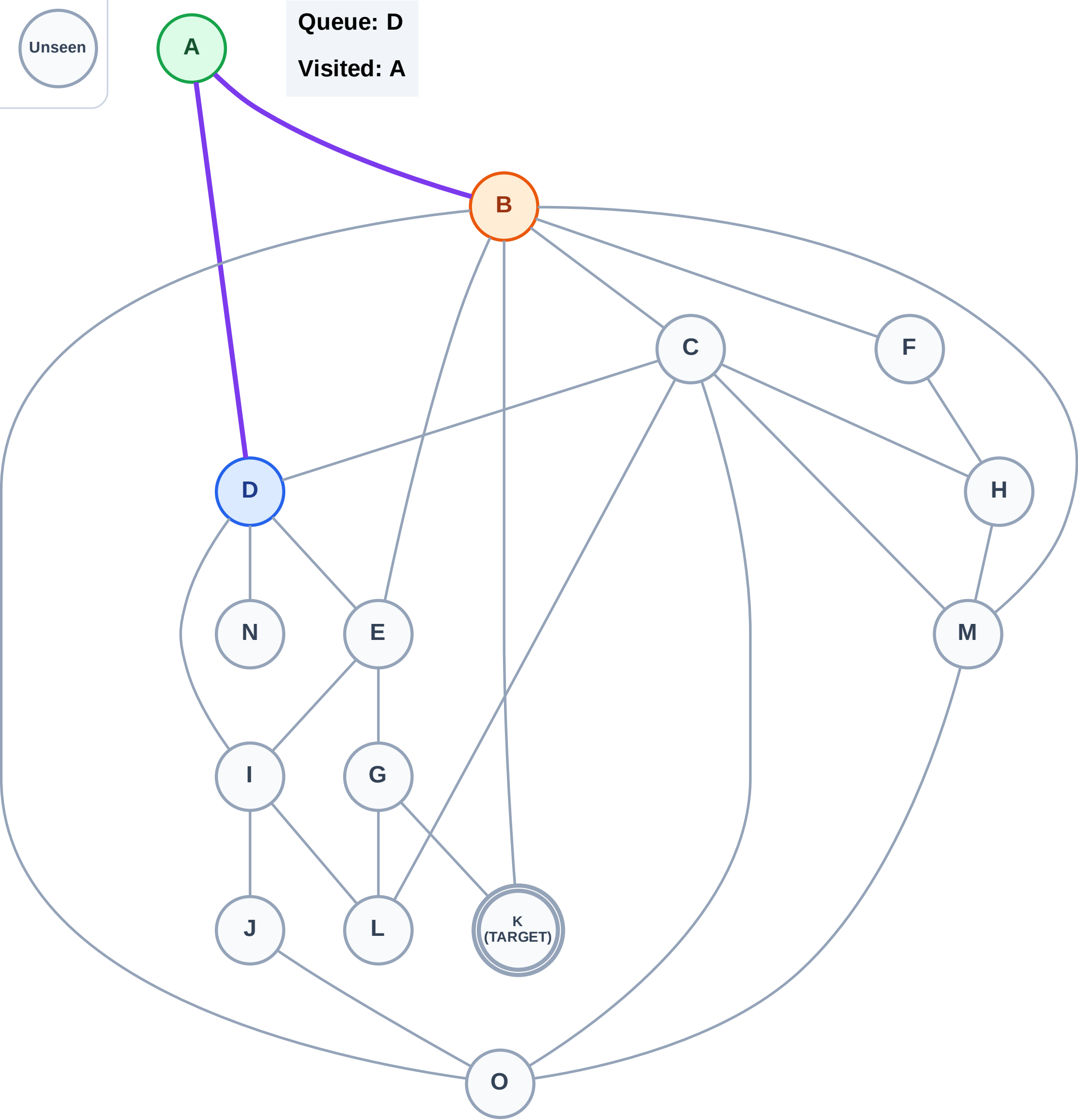
Legend

Complete

Processing

Discovered

Unseen



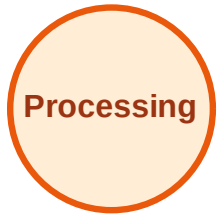
BFS Traversal

Step 5: Discover C, E, F, K, M, O from B

Legend



Complete



Processing



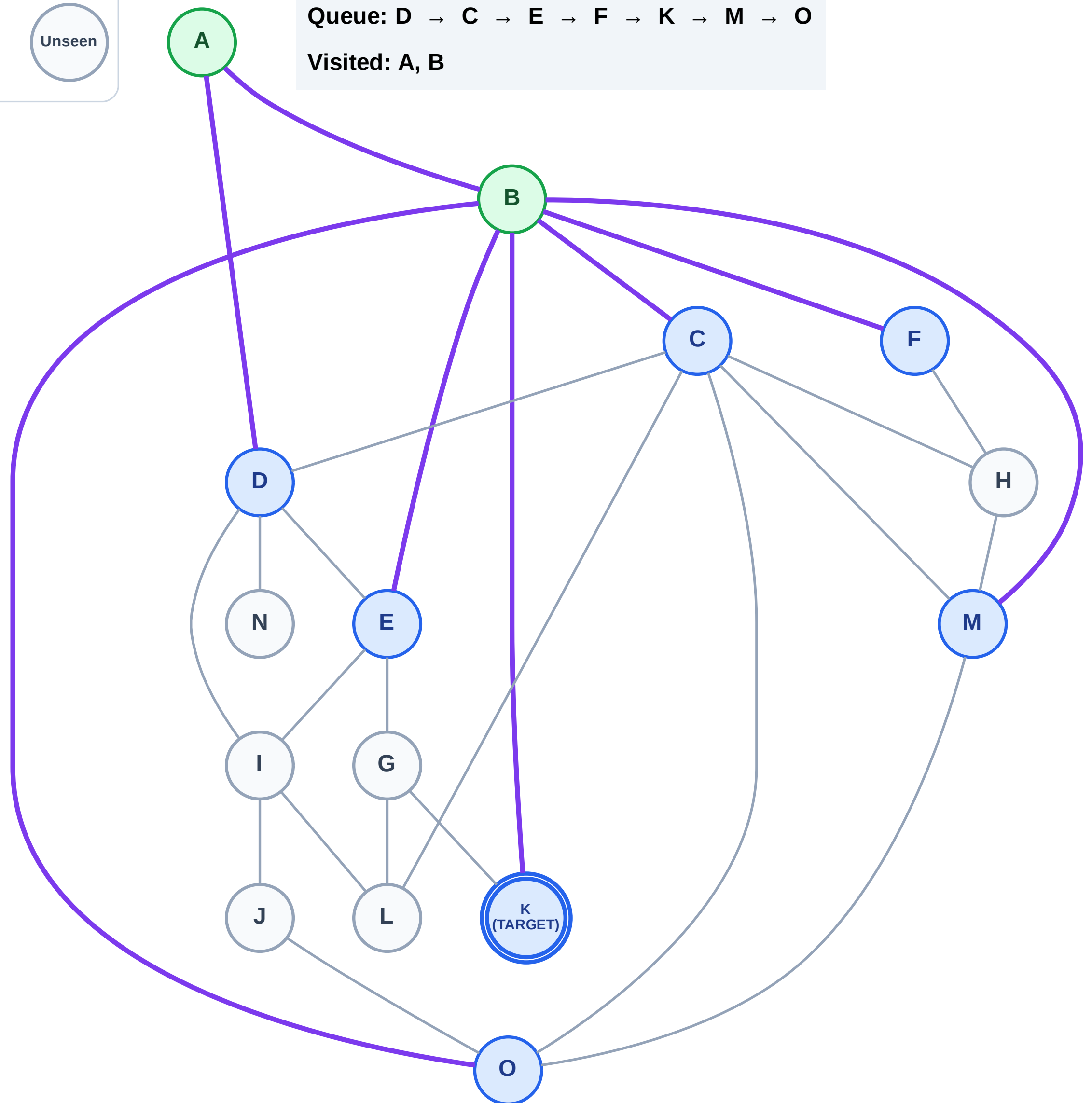
Discovered



Unseen

Queue: D → C → E → F → K → M → O

Visited: A, B



BFS Traversal

Step 6: Process node D

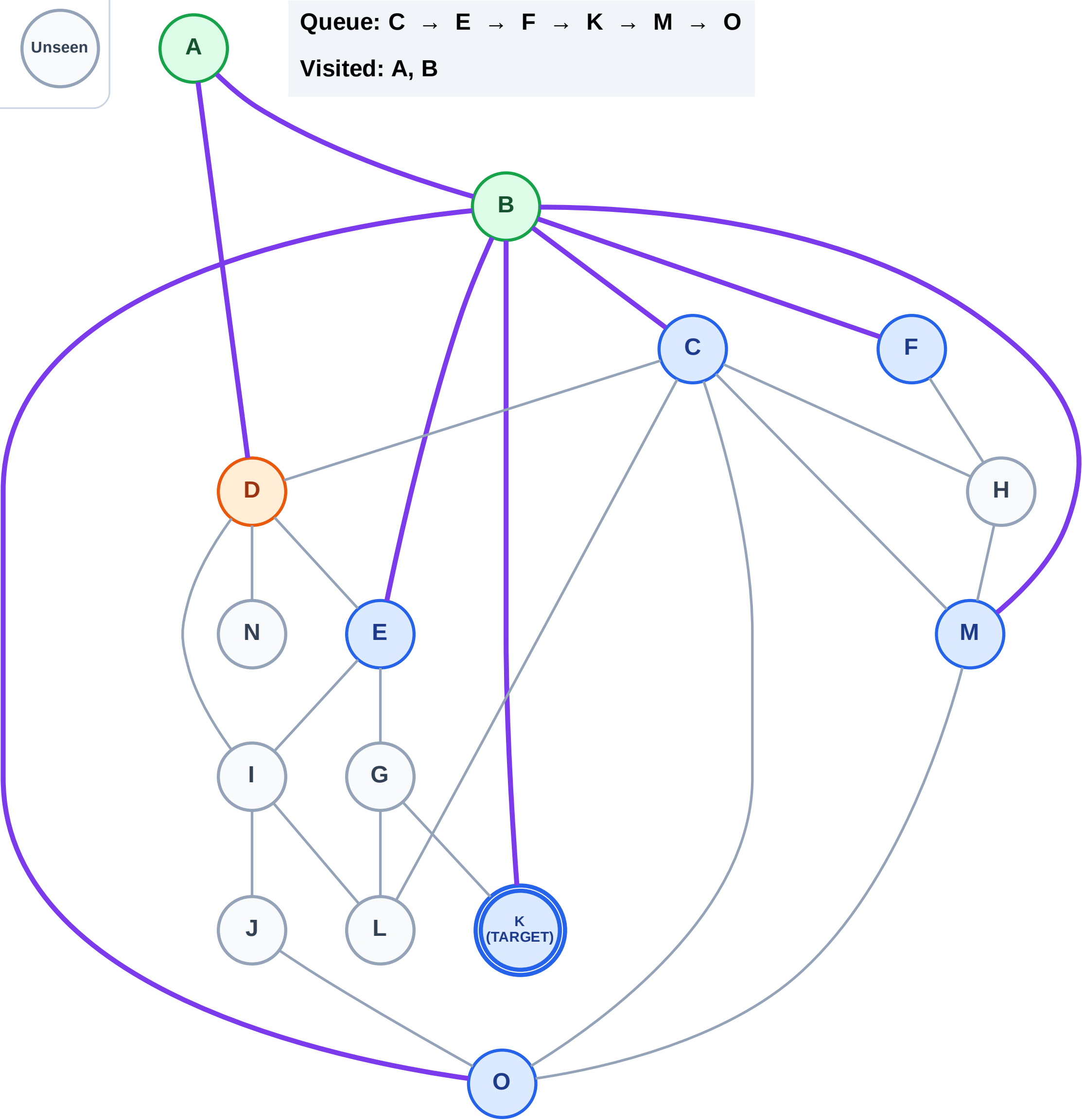
Legend

Complete

Processing

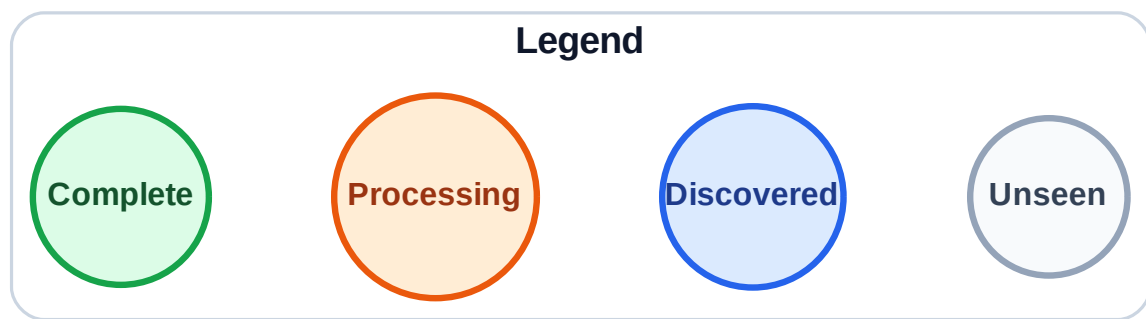
Discovered

Unseen



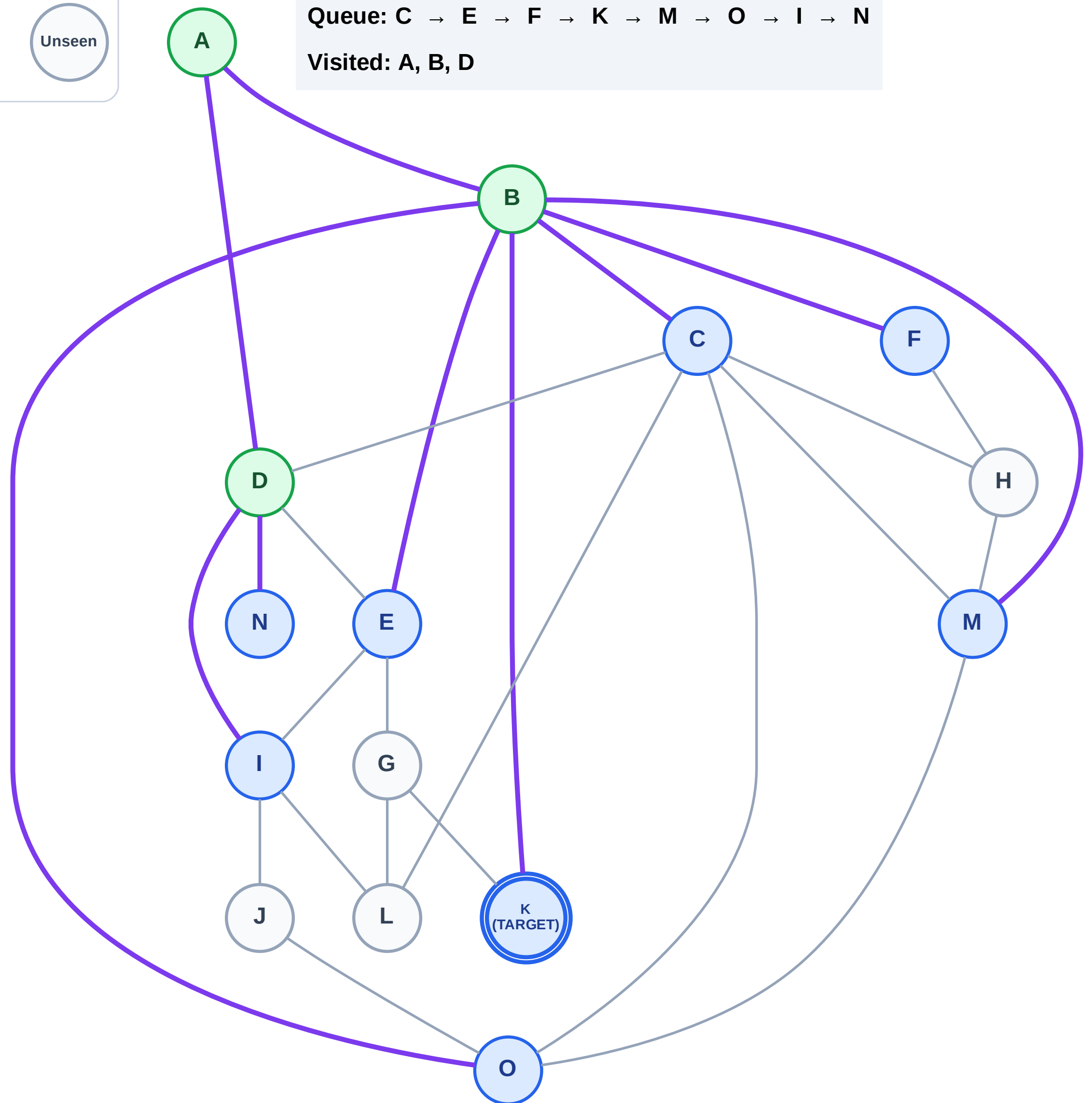
BFS Traversal

Step 7: Discover I, N from D



Queue: C → E → F → K → M → O → I → N

Visited: A, B, D



BFS Traversal

Step 8: Process node C

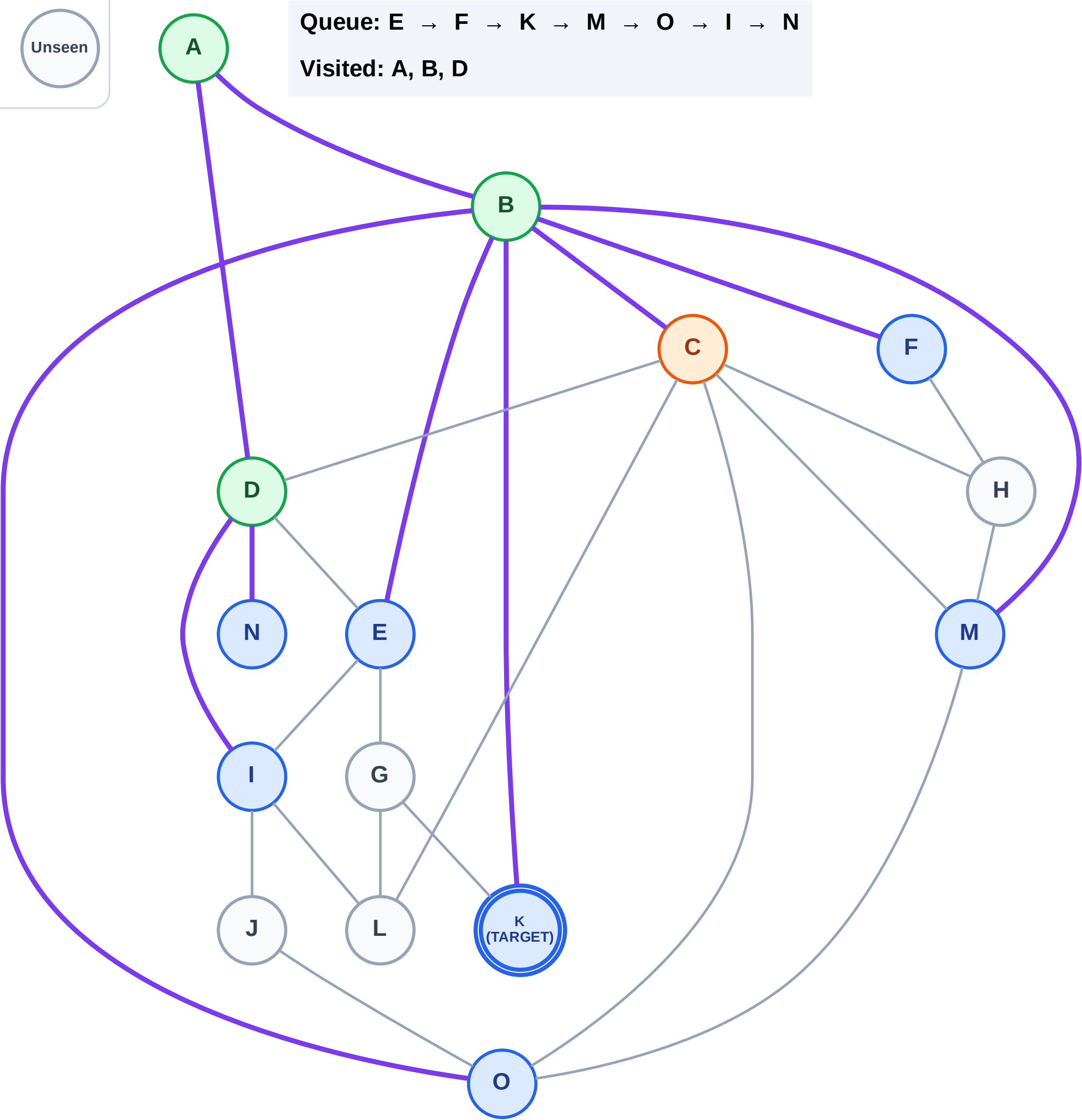
Legend

Complete

Processing

Discovered

Unseen



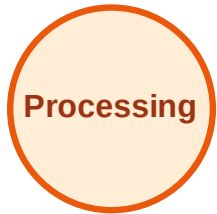
BFS Traversal

Step 9: Discover H, L from C

Legend



Complete



Processing



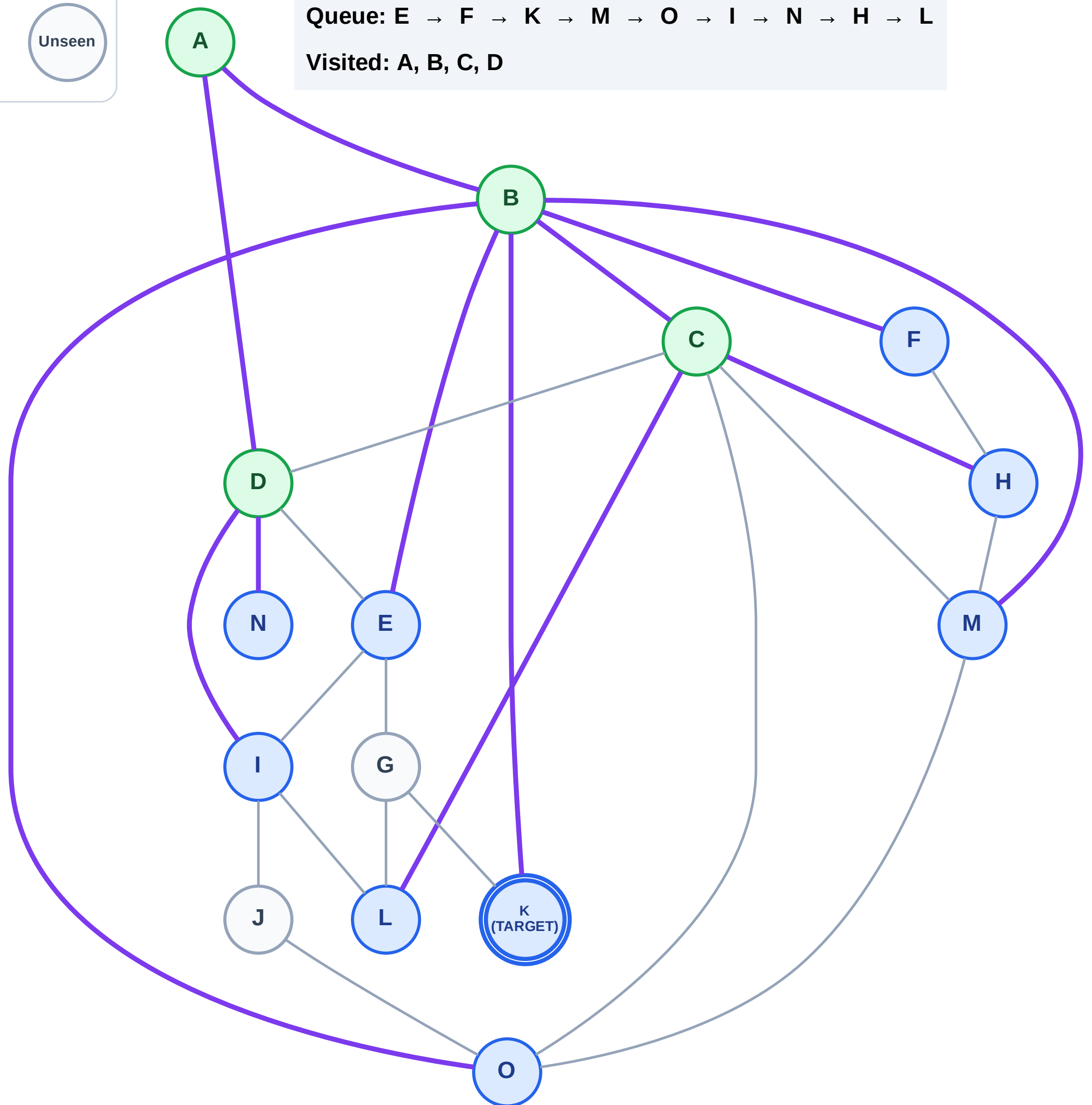
Discovered



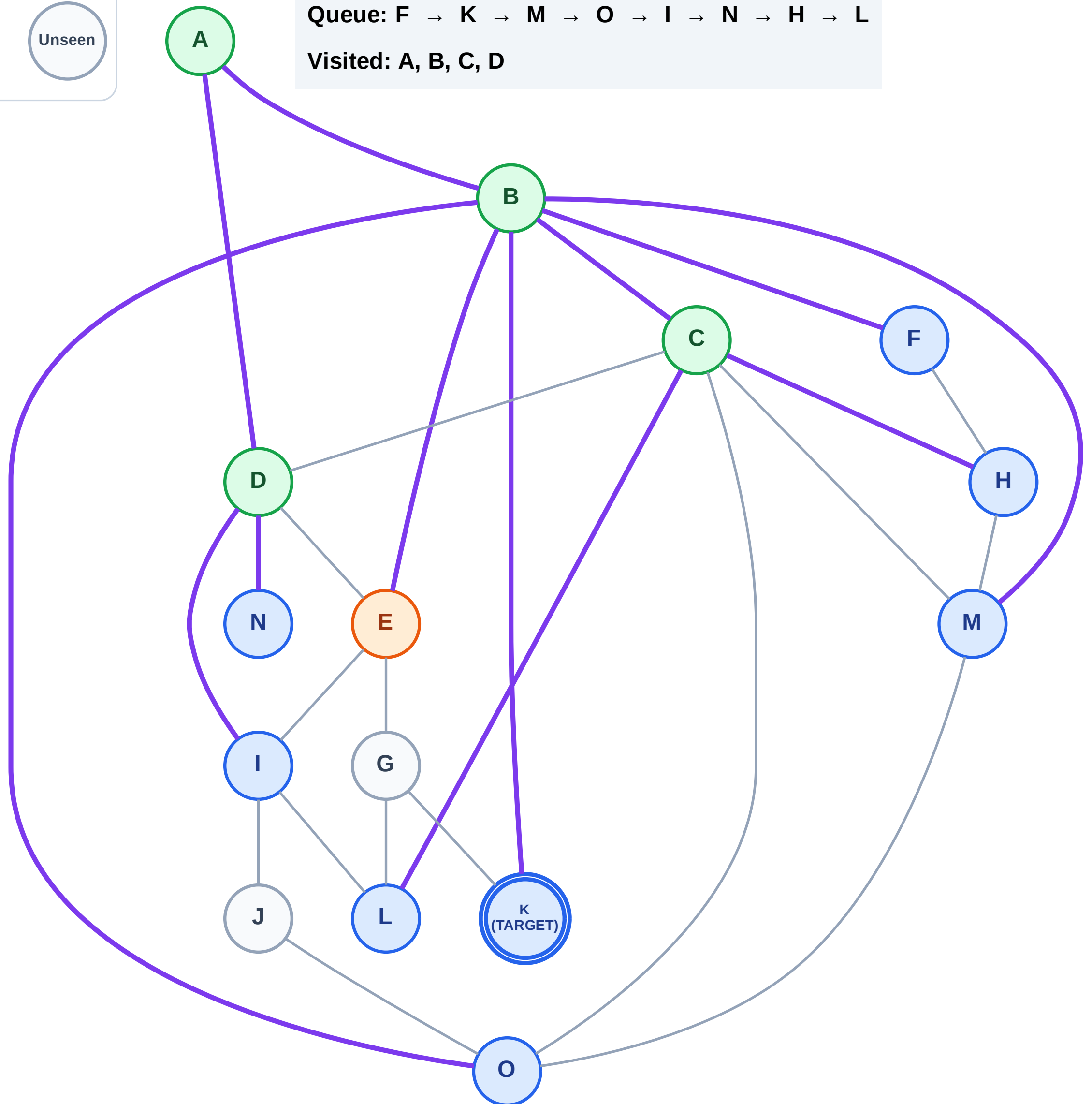
Unseen

Queue: E → F → K → M → O → I → N → H → L

Visited: A, B, C, D

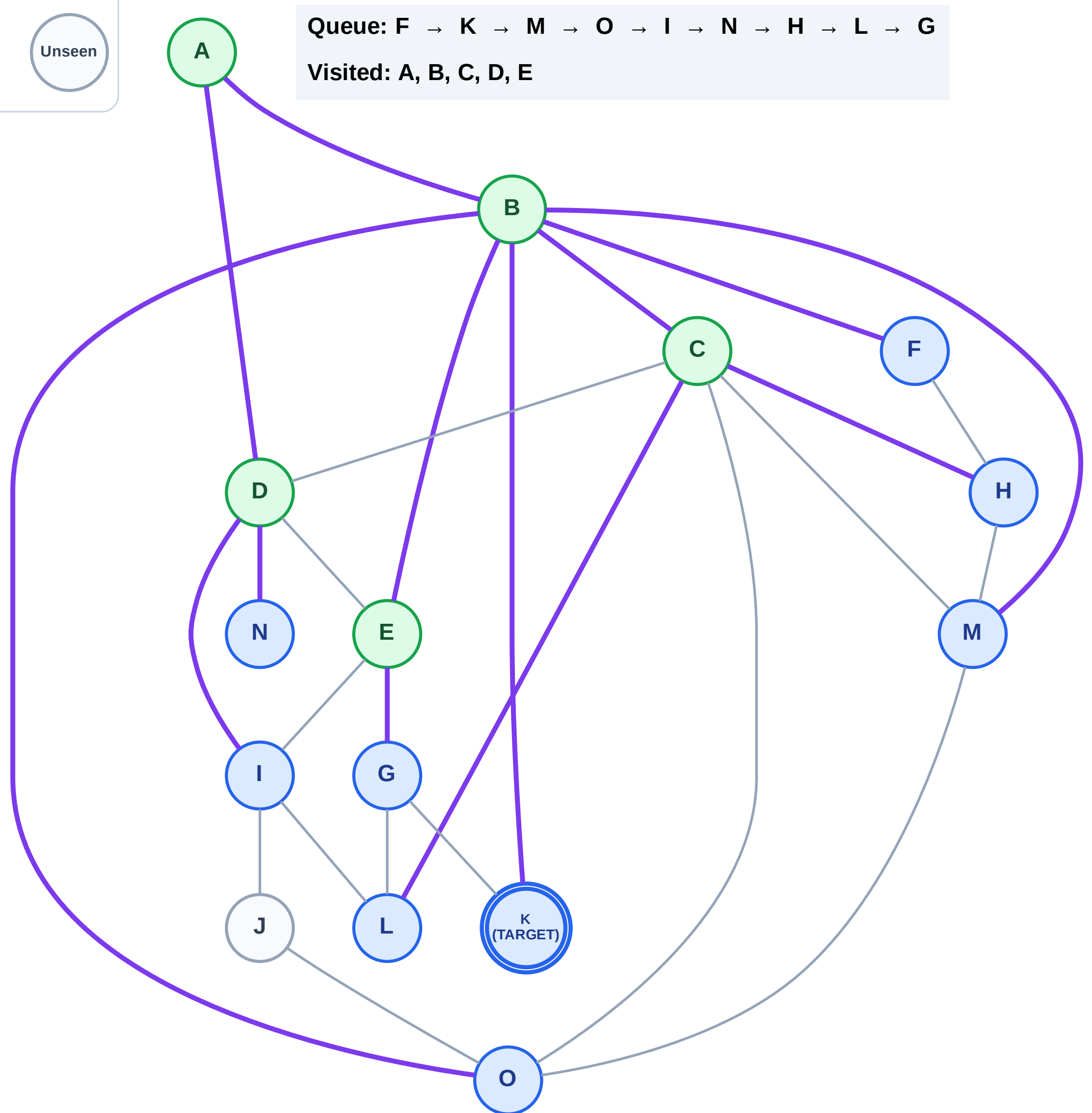
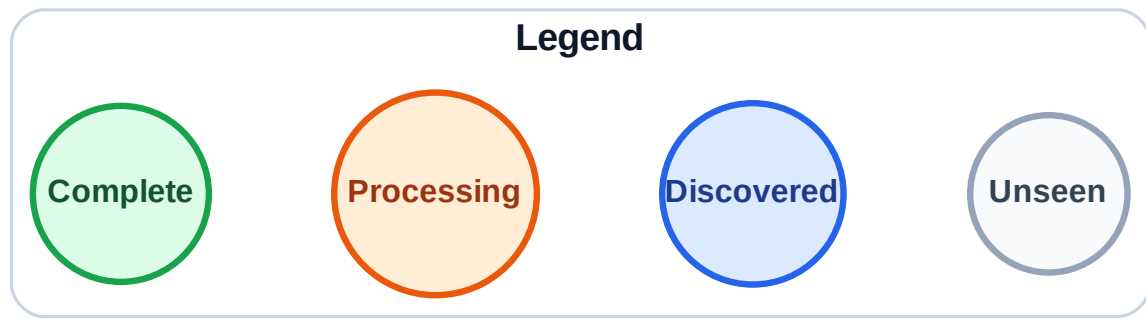


Step 10: Process node E

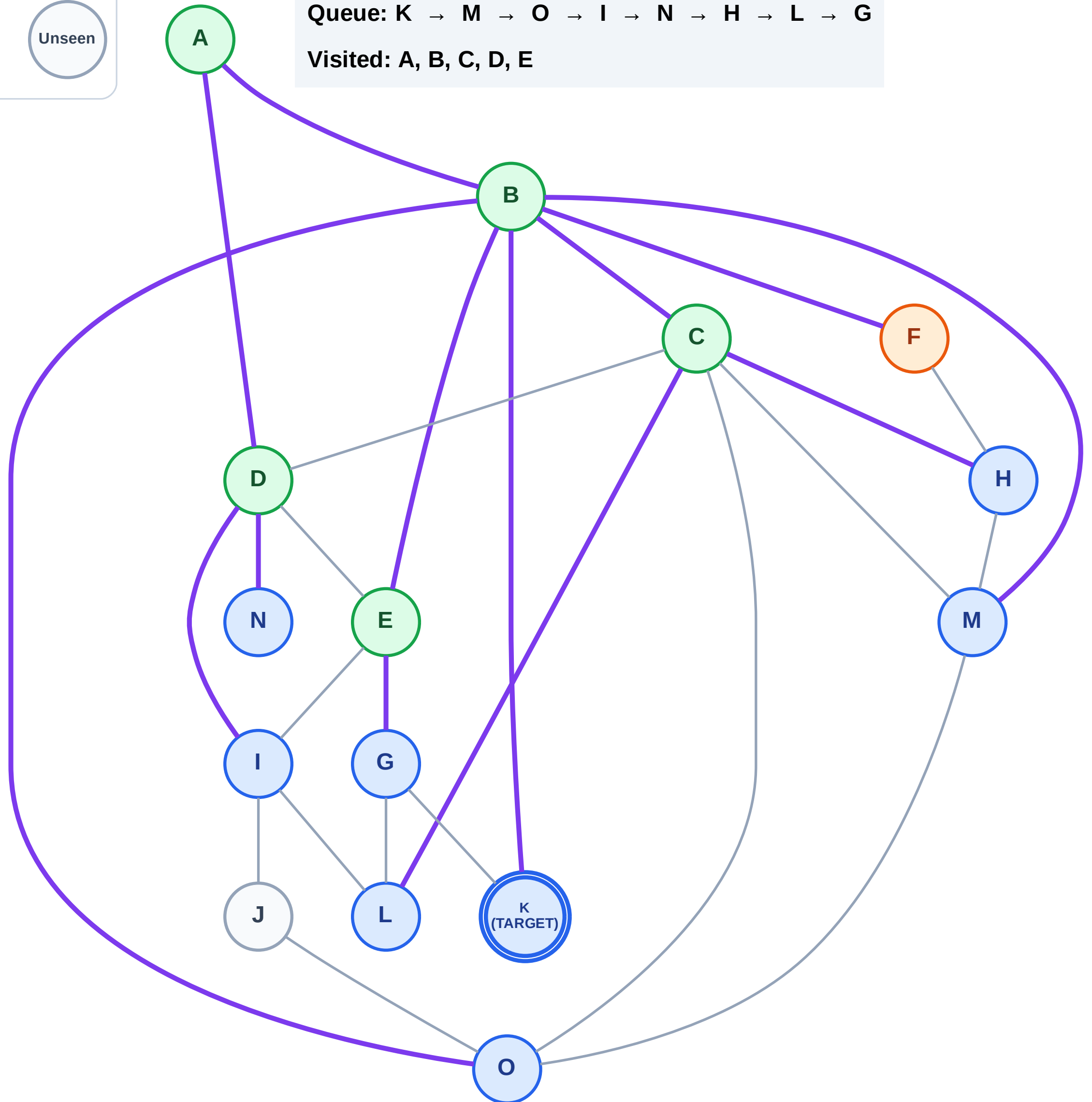


BFS Traversal

Step 11: Discover G from E



Step 12: Process node F



BFS Traversal

Step 13: No new neighbors from F

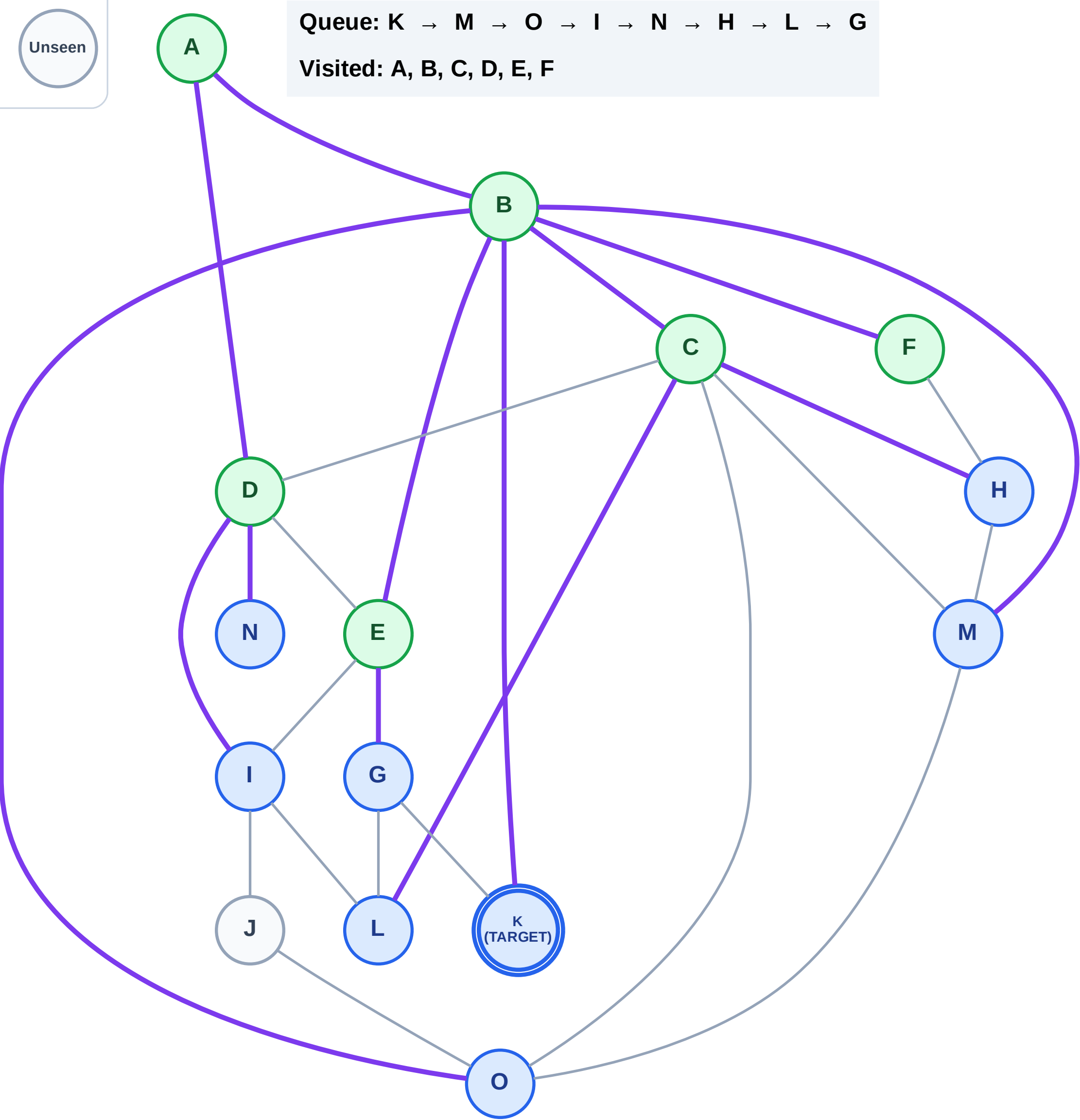
Legend

Complete

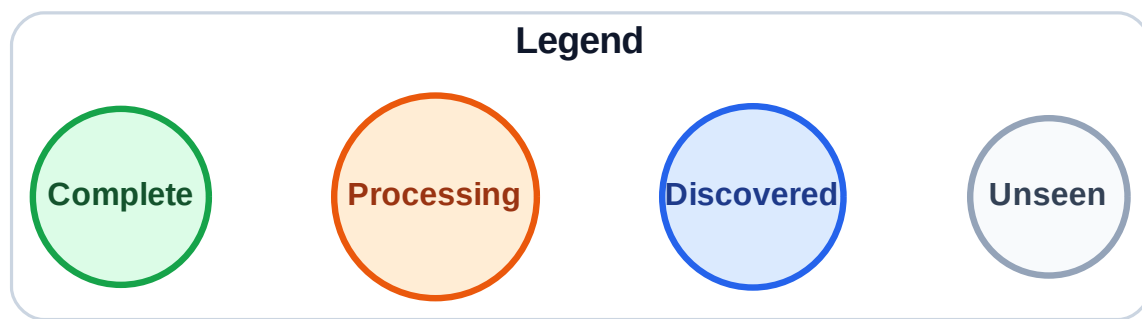
Processing

Discovered

Unseen

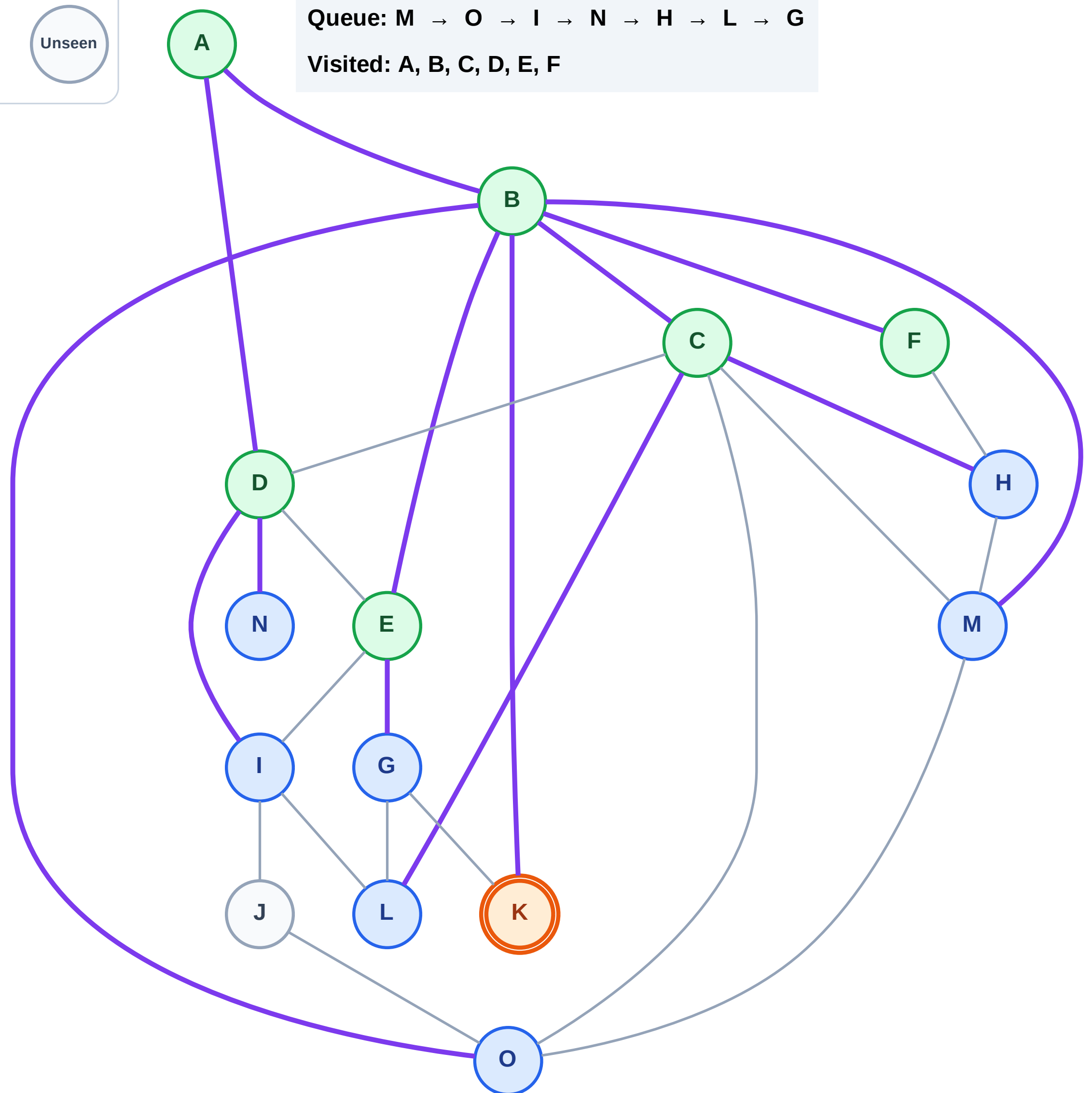


Step 14: Process node K



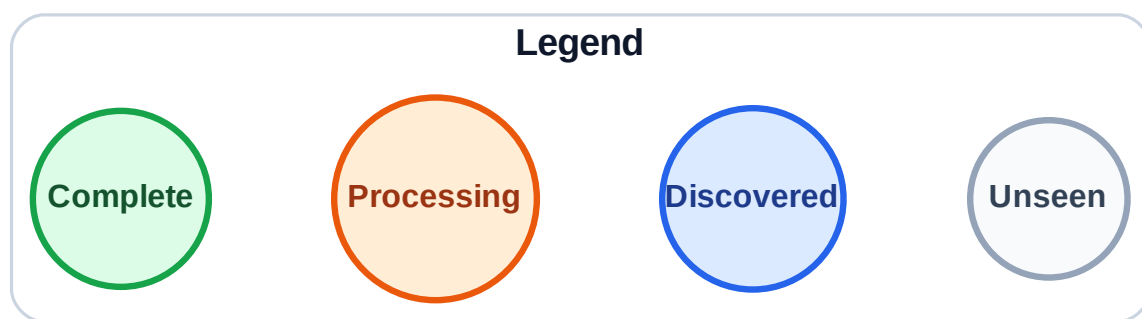
Queue: M → O → I → N → H → L → G

Visited: A, B, C, D, E, F



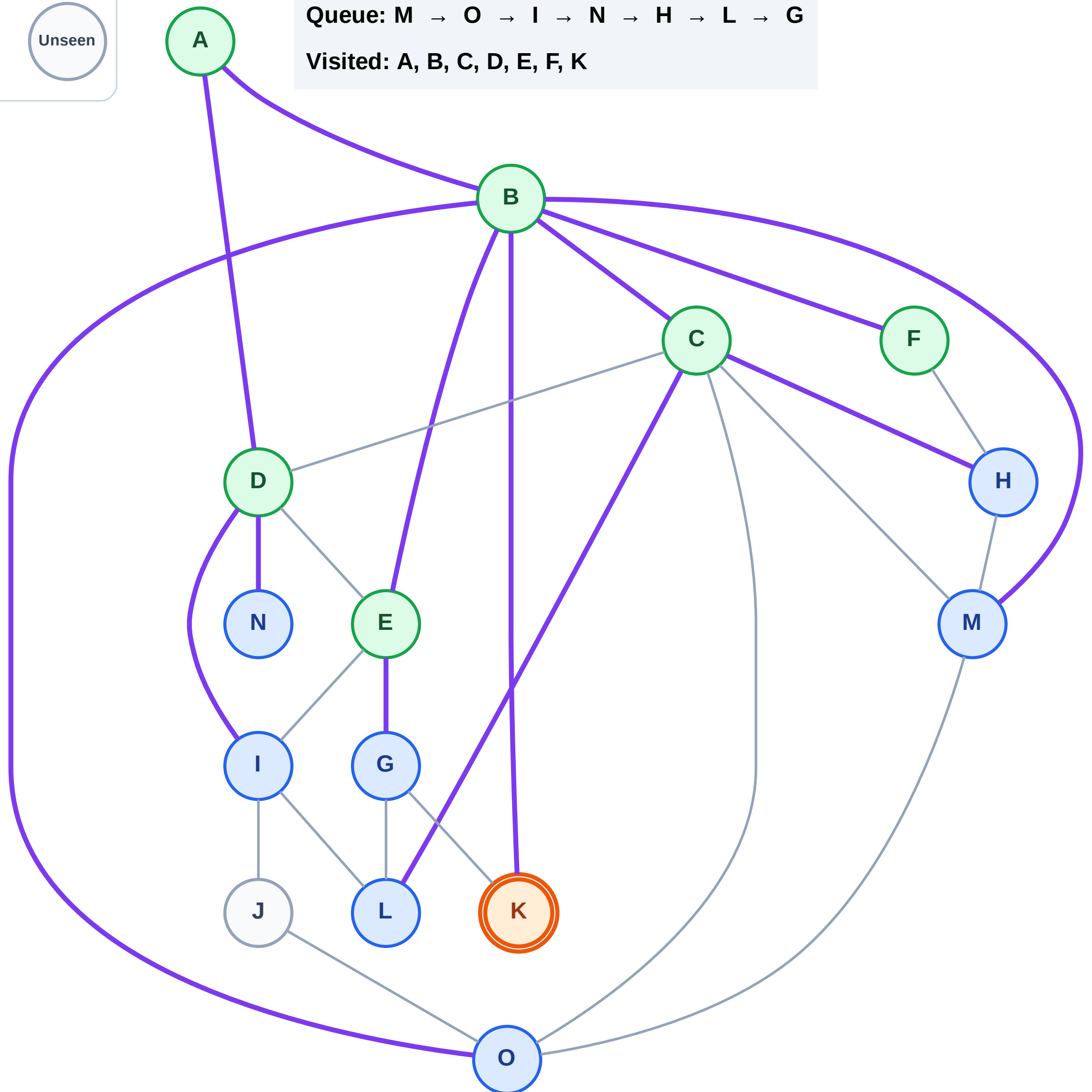
BFS Traversal

Step 15: TARGET NODE K REACHED! Path (2 hops): A → B → K



Queue: M → O → I → N → H → L → G

Visited: A, B, C, D, E, F, K



DFS Traversal

Step 1: Start at A (Seeking target node K)

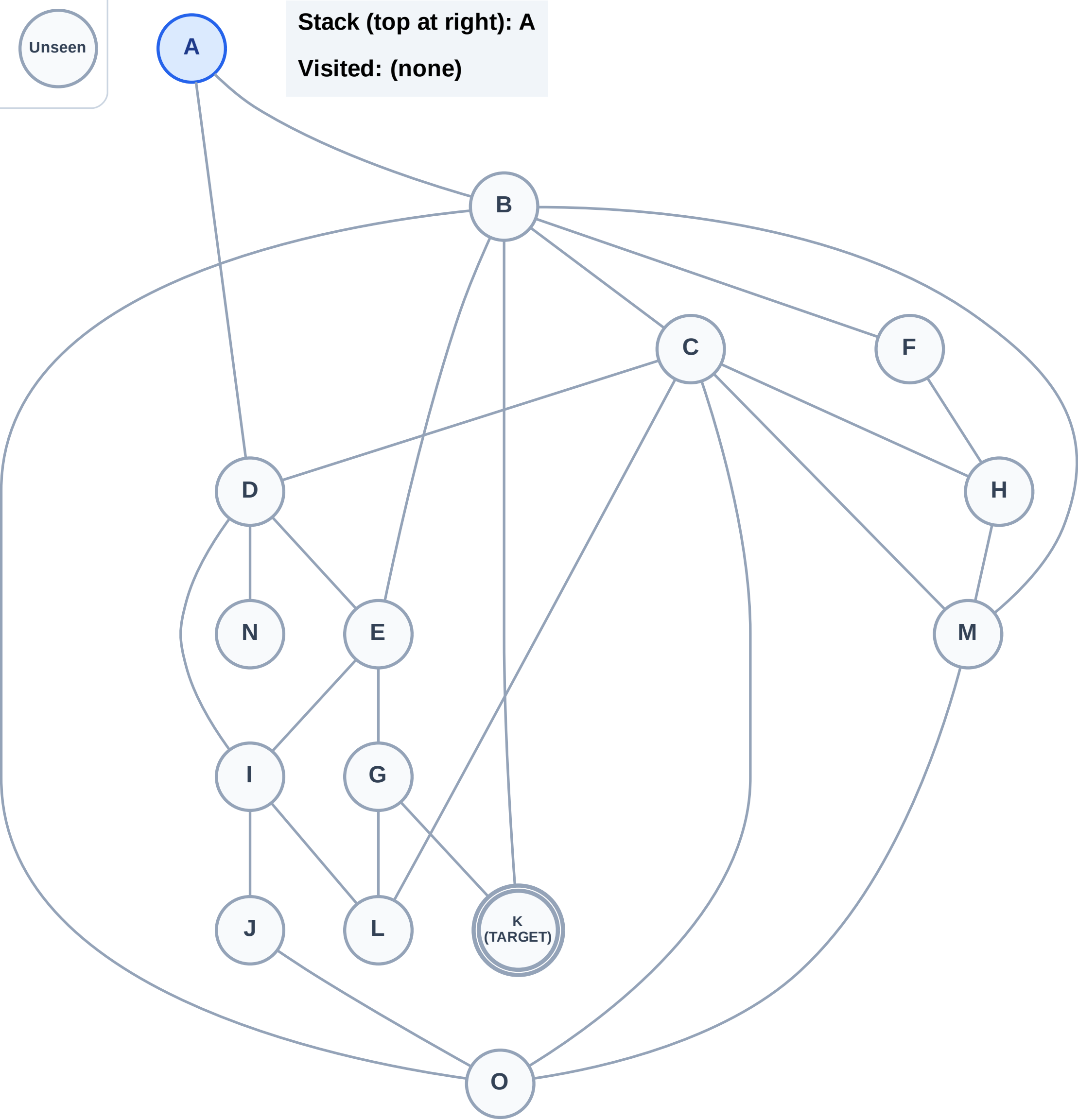
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 2: Process node A

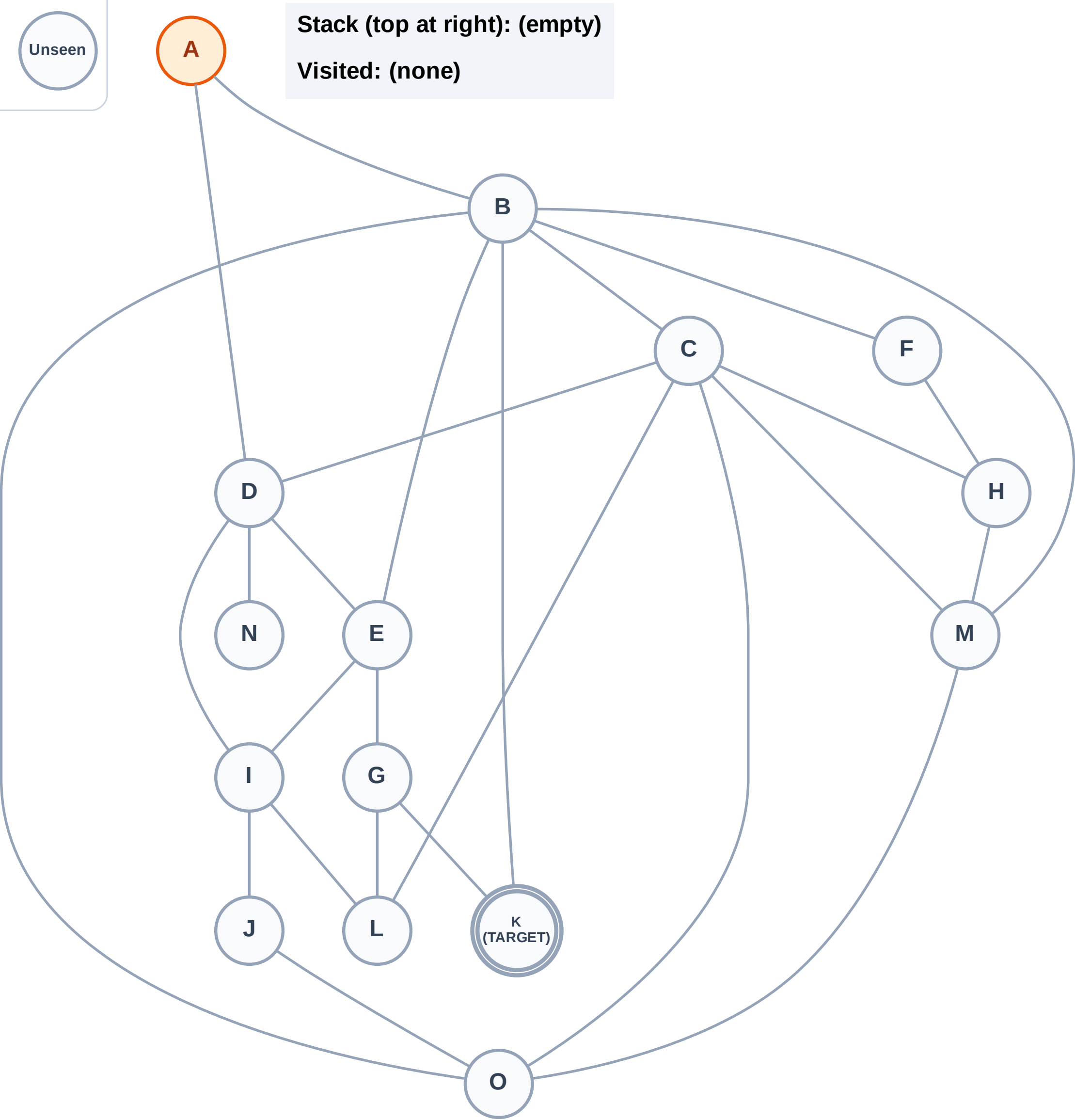
Legend

Complete

Processing

Discovered

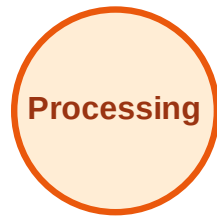
Unseen



DFS Traversal

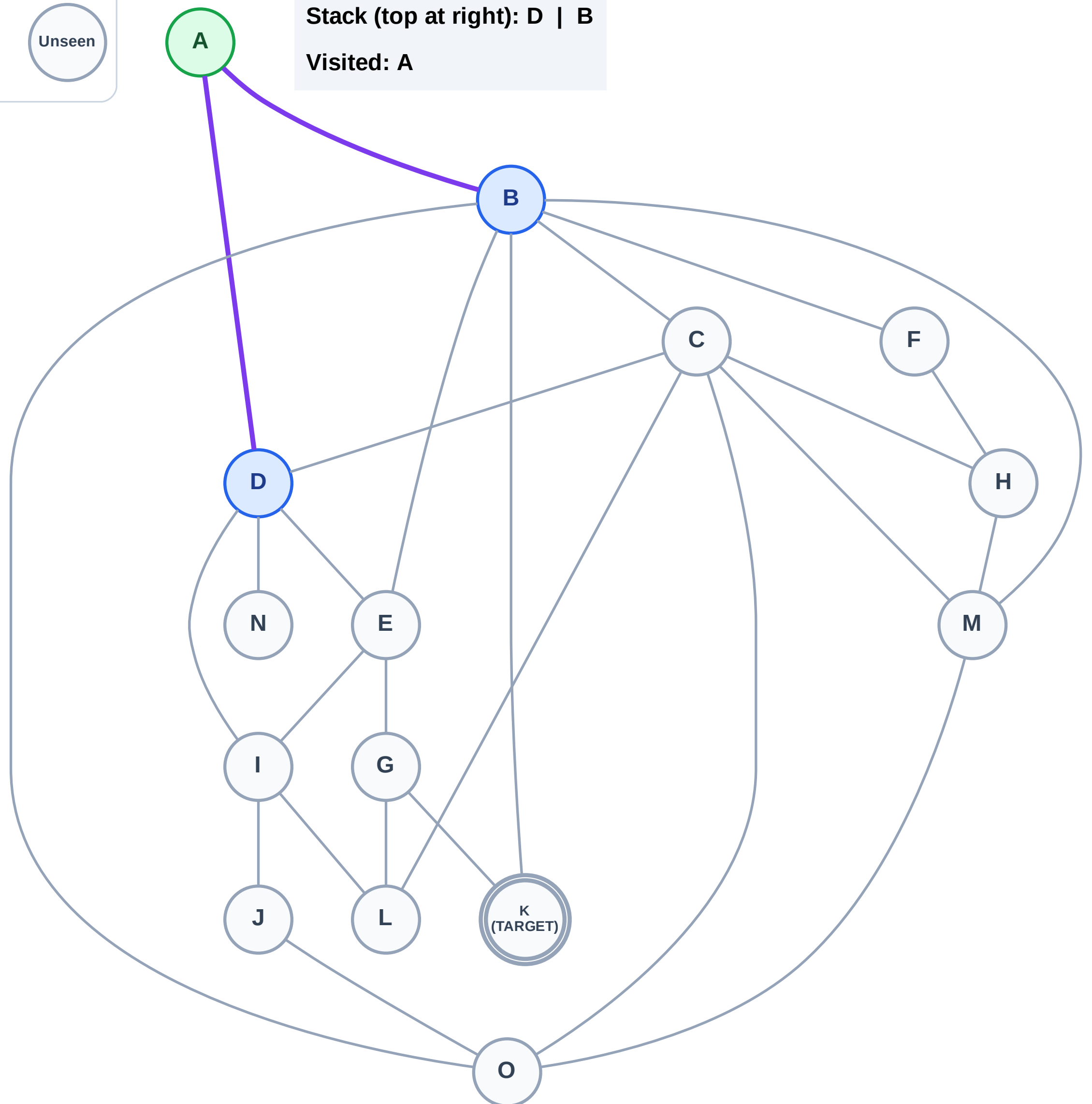
Step 3: Discover D, B from A

Legend



Stack (top at right): D | B

Visited: A



DFS Traversal

Step 4: Process node B

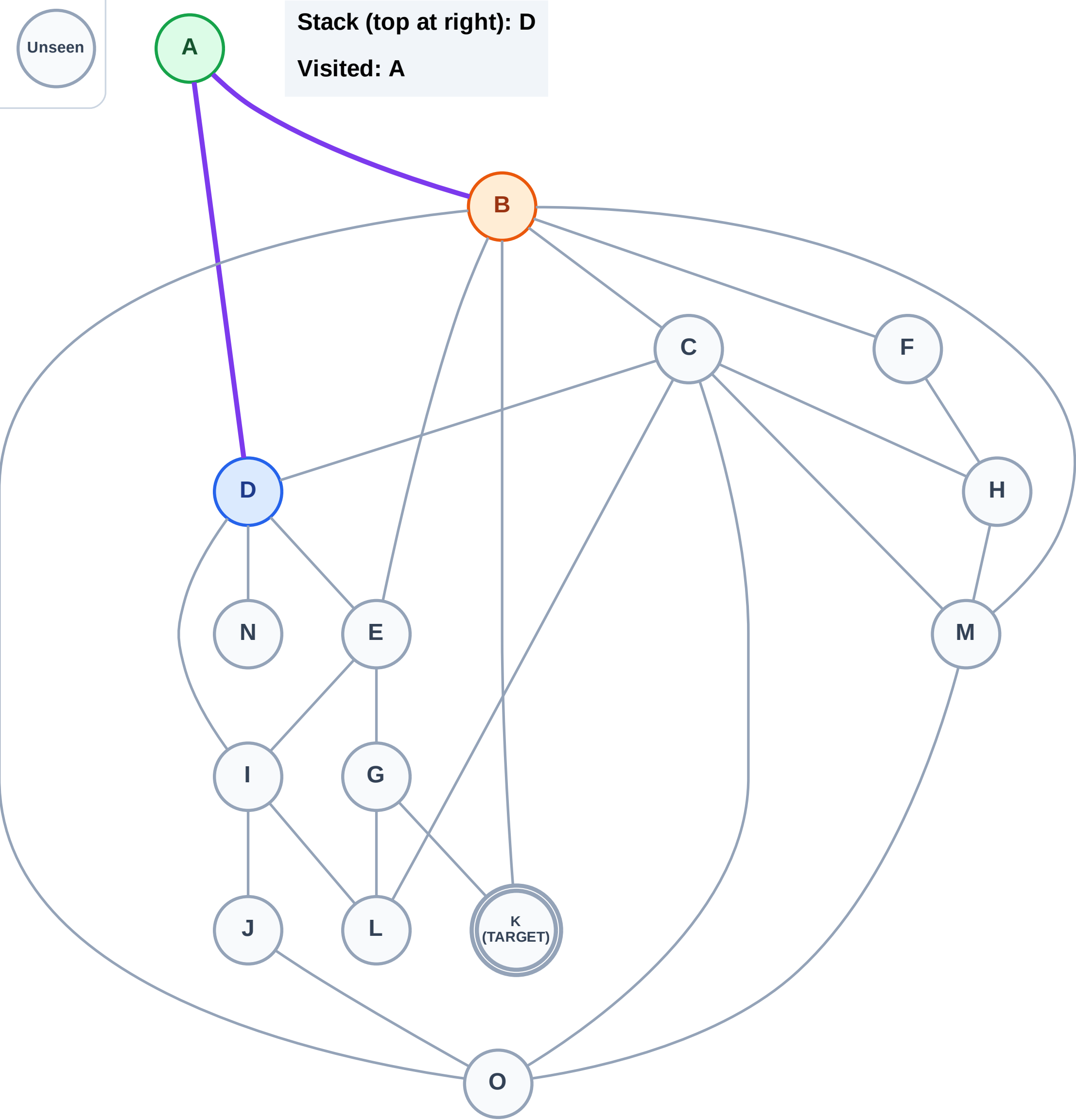
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 5: Discover O, M, K, F, E, C from B

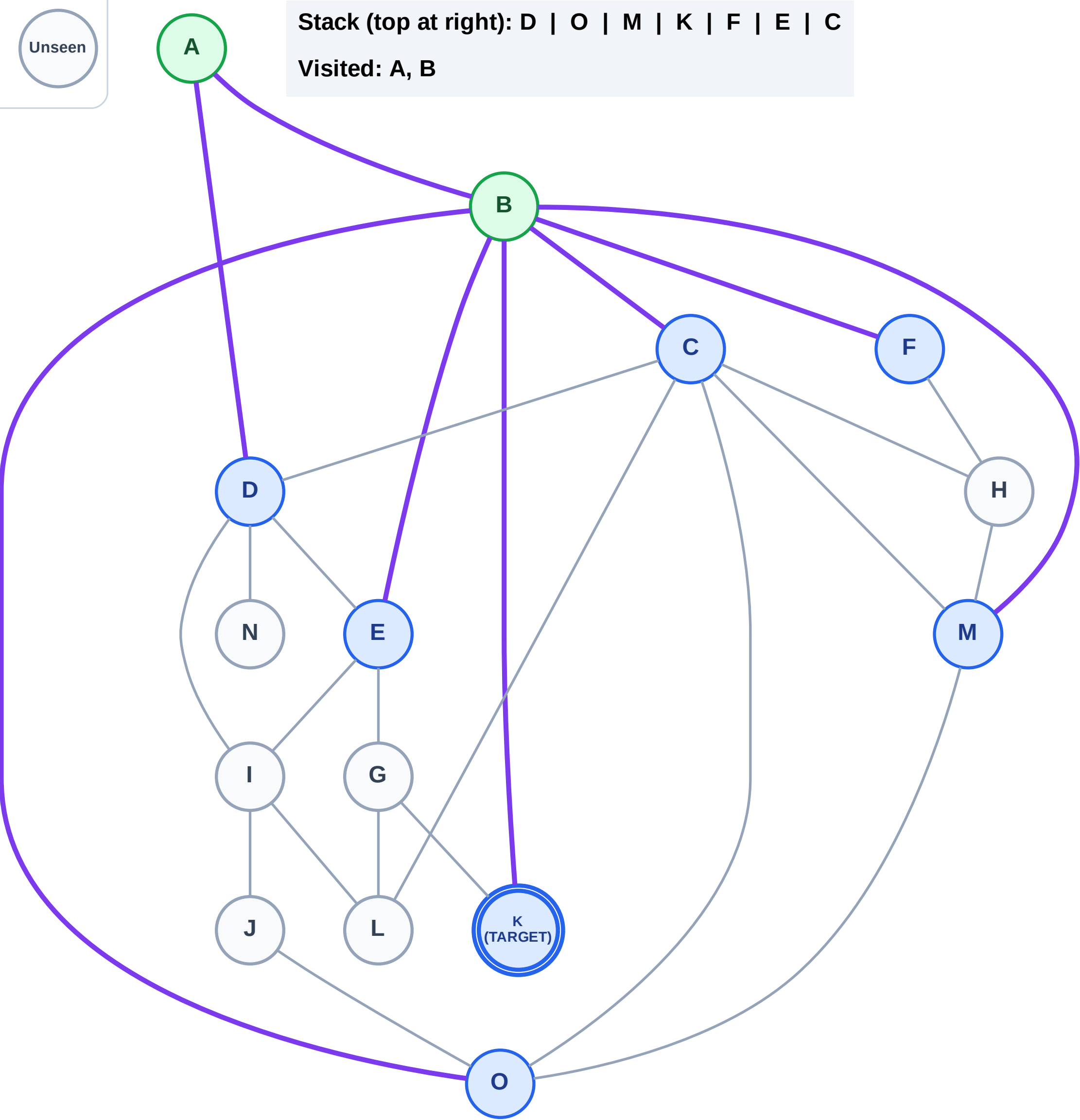
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 6: Process node C

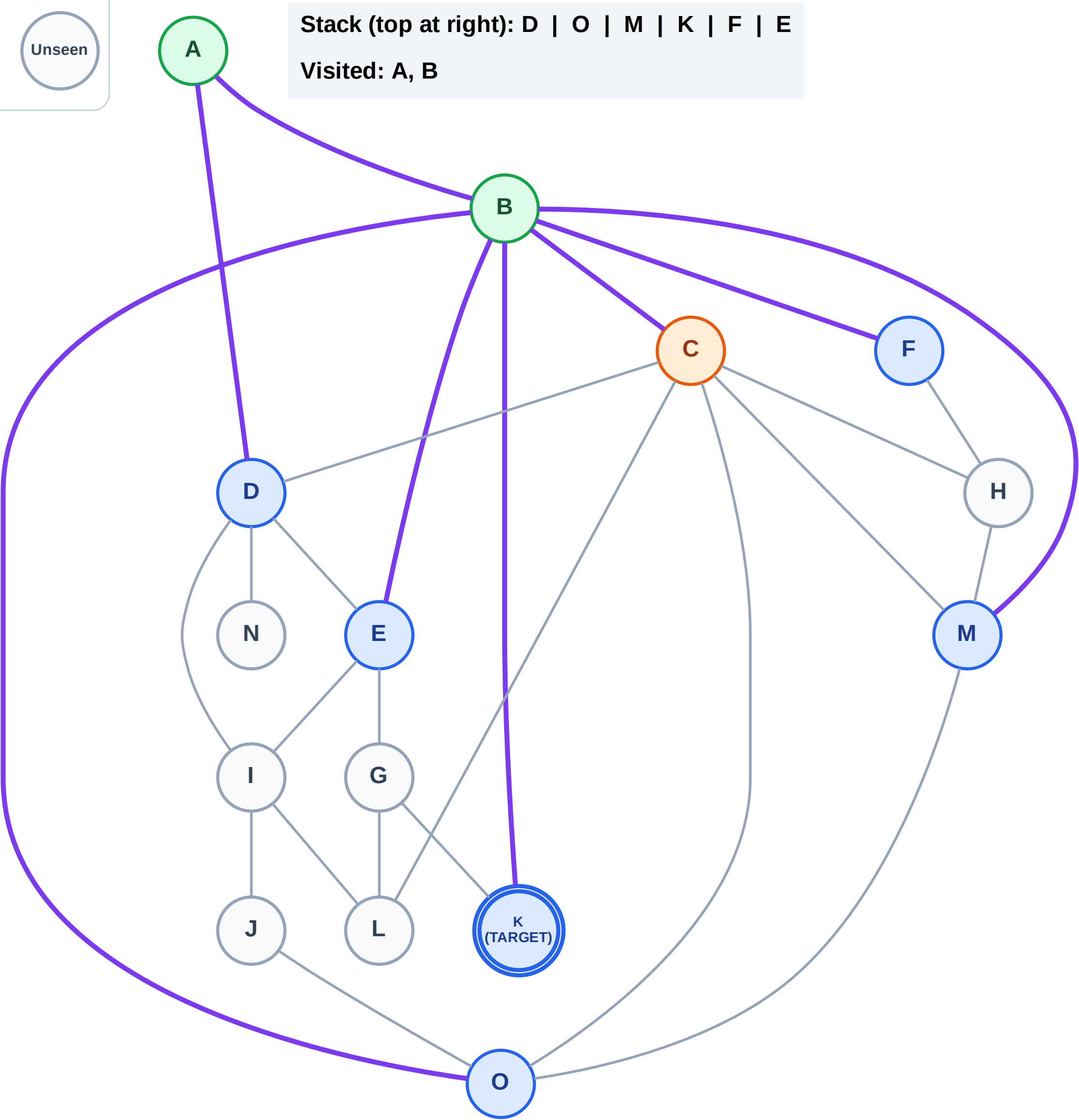
Legend

Complete

Processing

Discovered

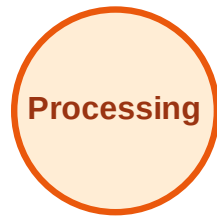
Unseen



DFS Traversal

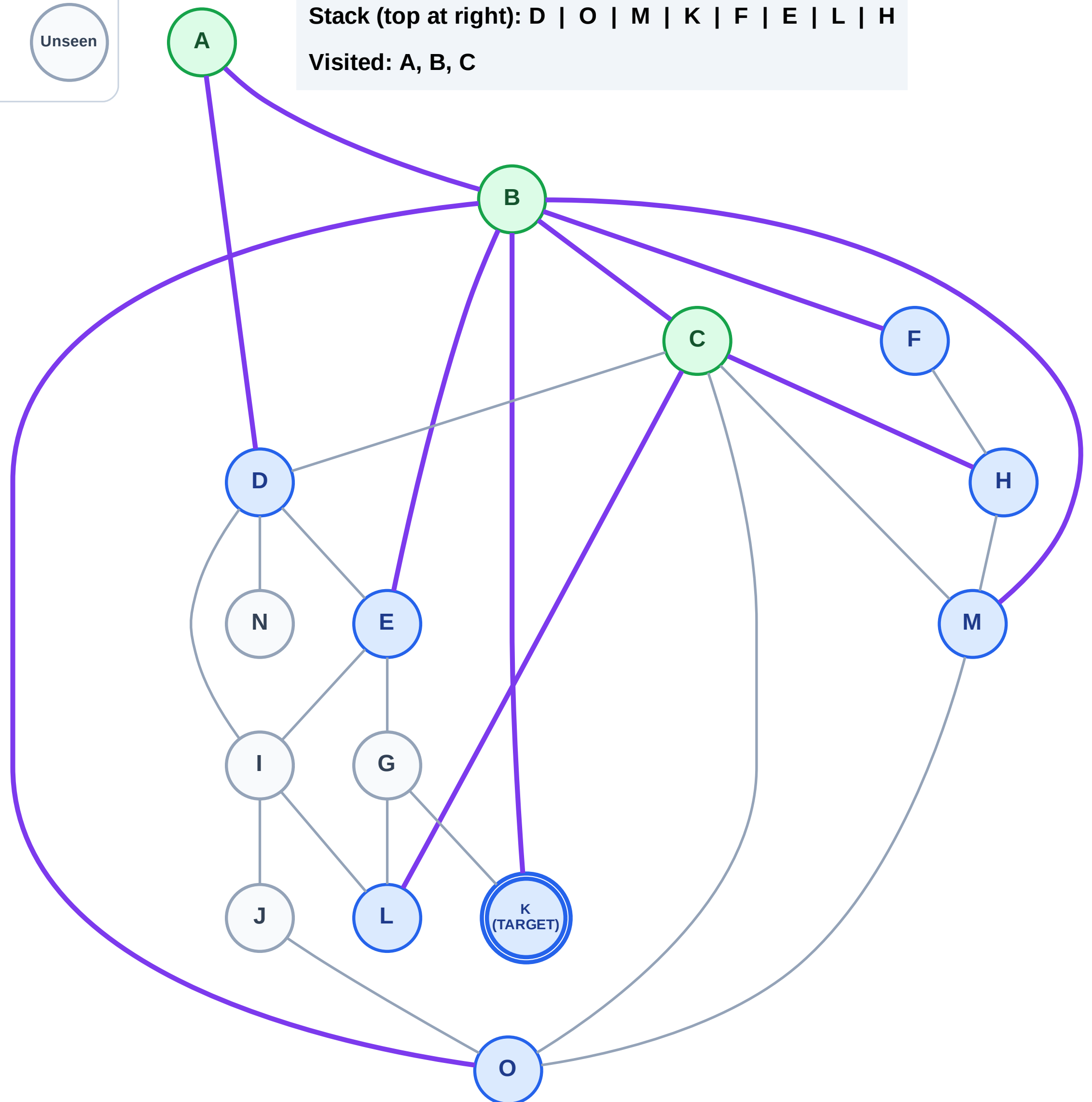
Step 7: Discover L, H from C

Legend



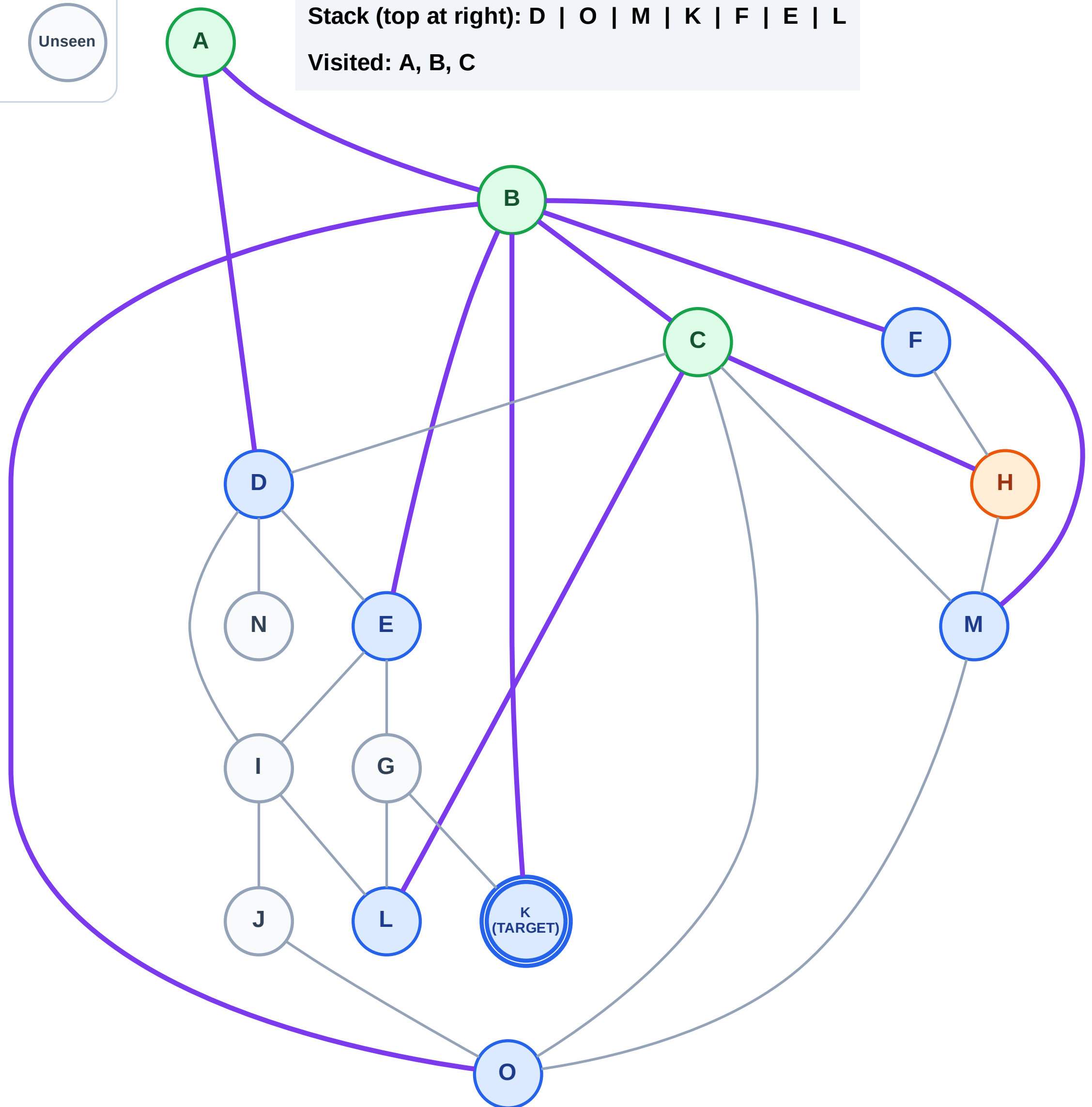
Stack (top at right): D | O | M | K | F | E | L | H

Visited: A, B, C



Step 8: Process node H

Step 8: Process node H



DFS Traversal

Step 9: No new neighbors from H

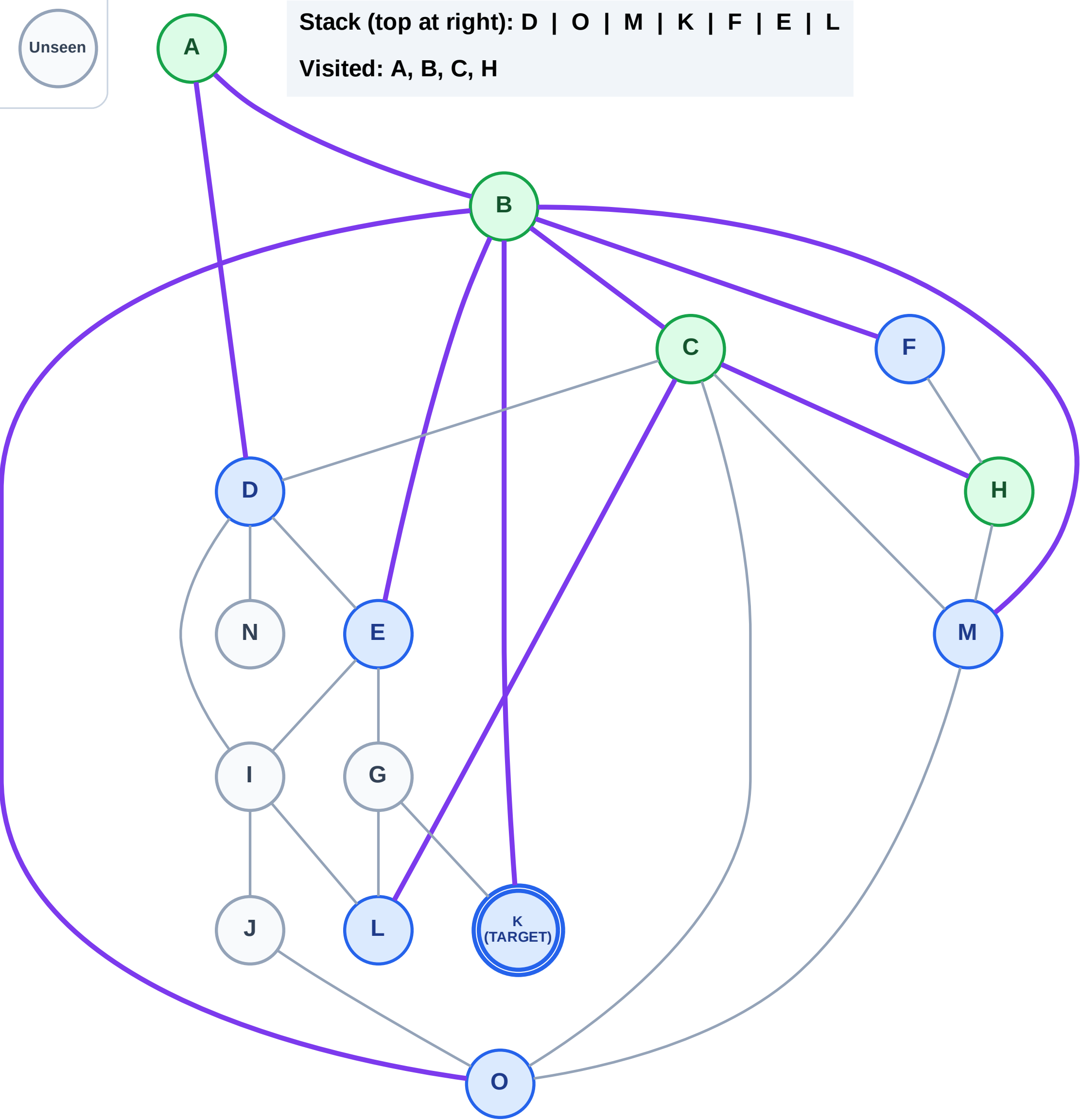
Legend

Complete

Processing

Discovered

Unseen



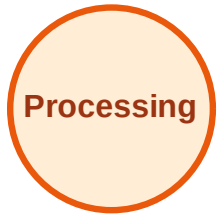
DFS Traversal

Step 10: Process node L

Legend



Complete



Processing



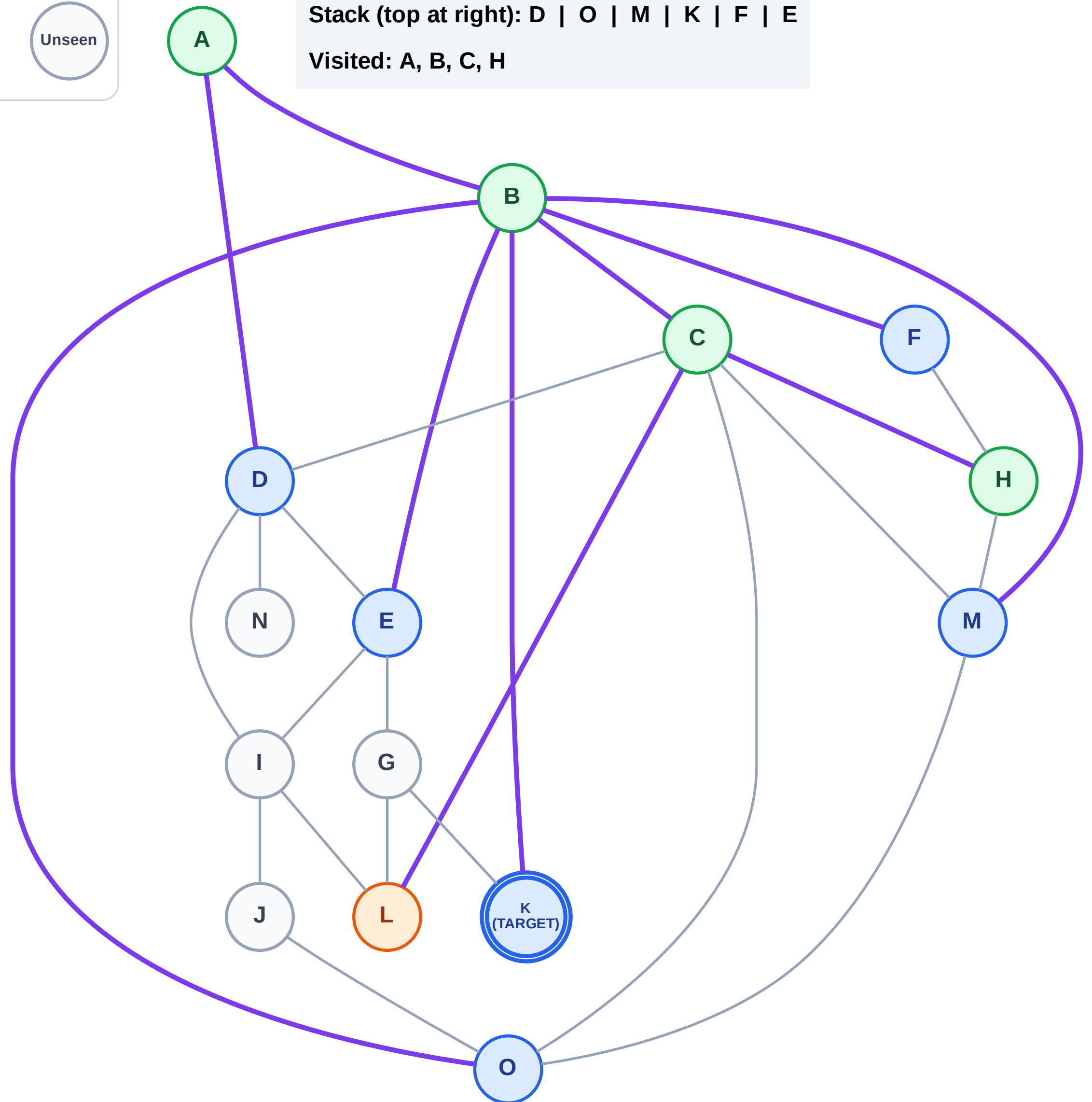
Discovered



Unseen

Stack (top at right): D | O | M | K | F | E

Visited: A, B, C, H



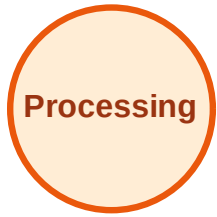
DFS Traversal

Step 11: Discover I, G from L

Legend



Complete



Processing



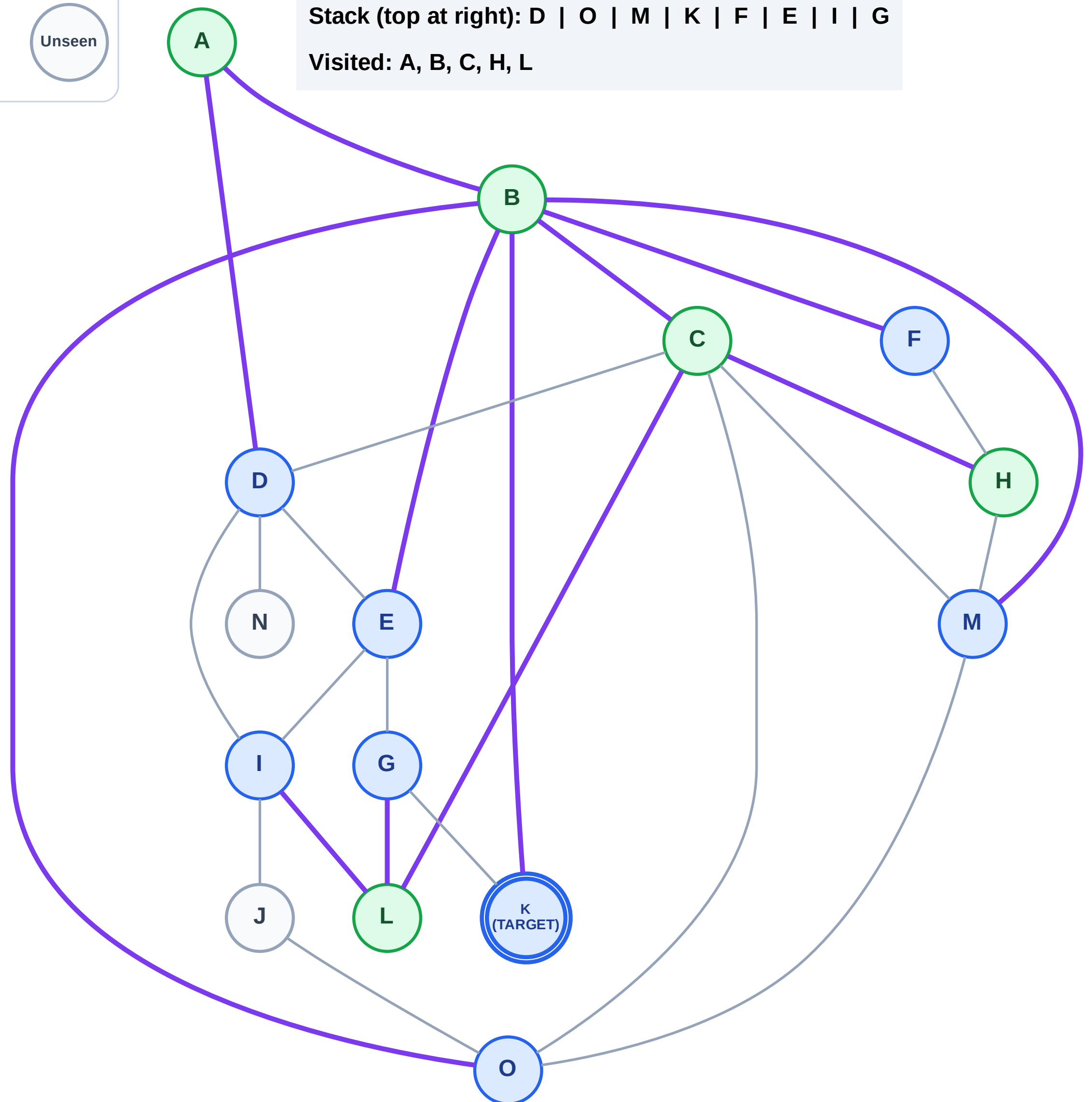
Discovered



Unseen

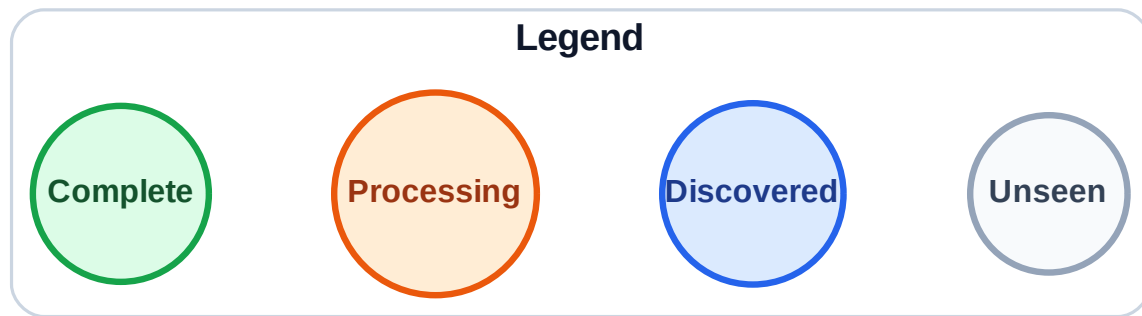
Stack (top at right): D | O | M | K | F | E | I | G

Visited: A, B, C, H, L

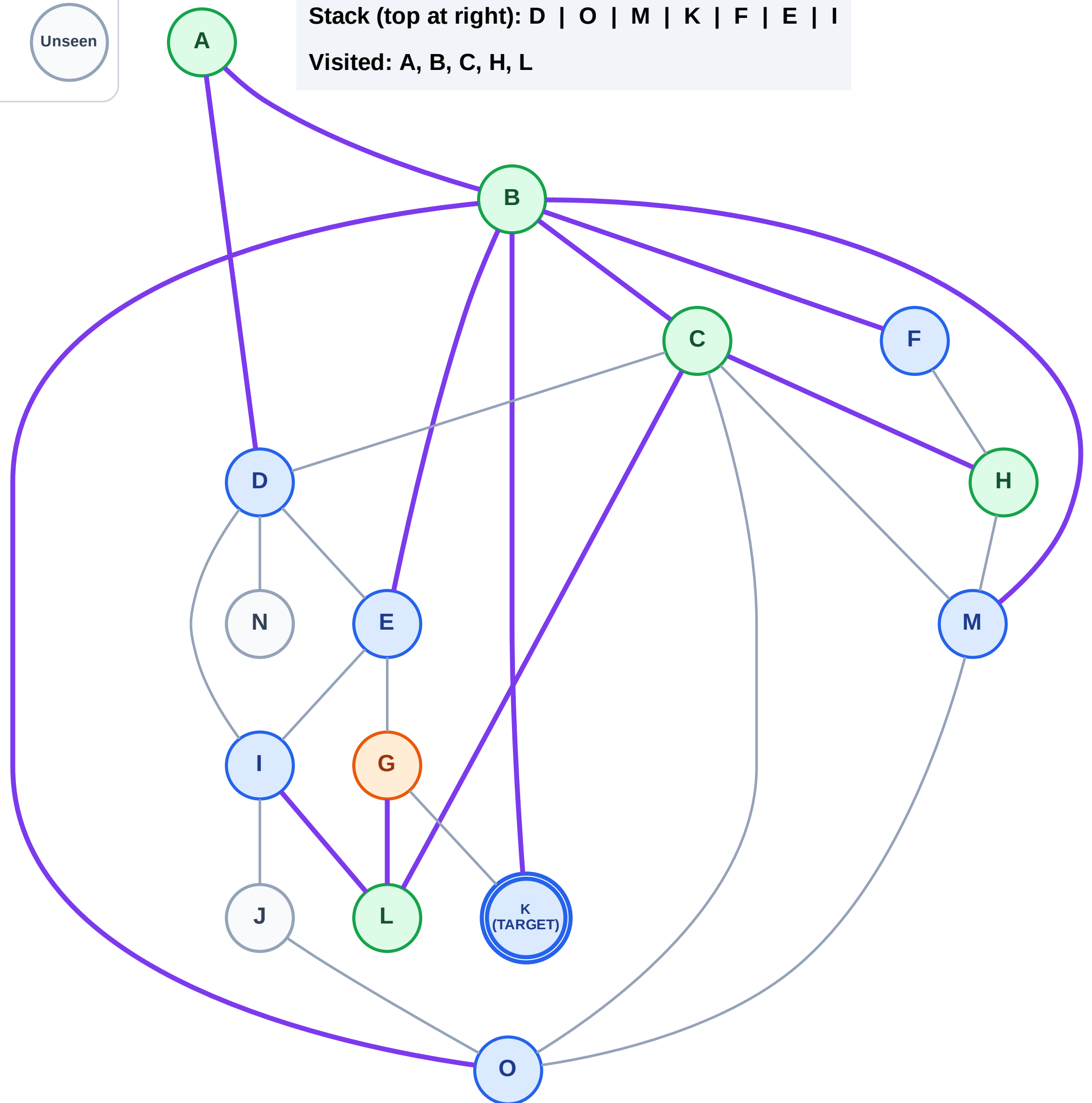


DFS Traversal

Step 12: Process node G

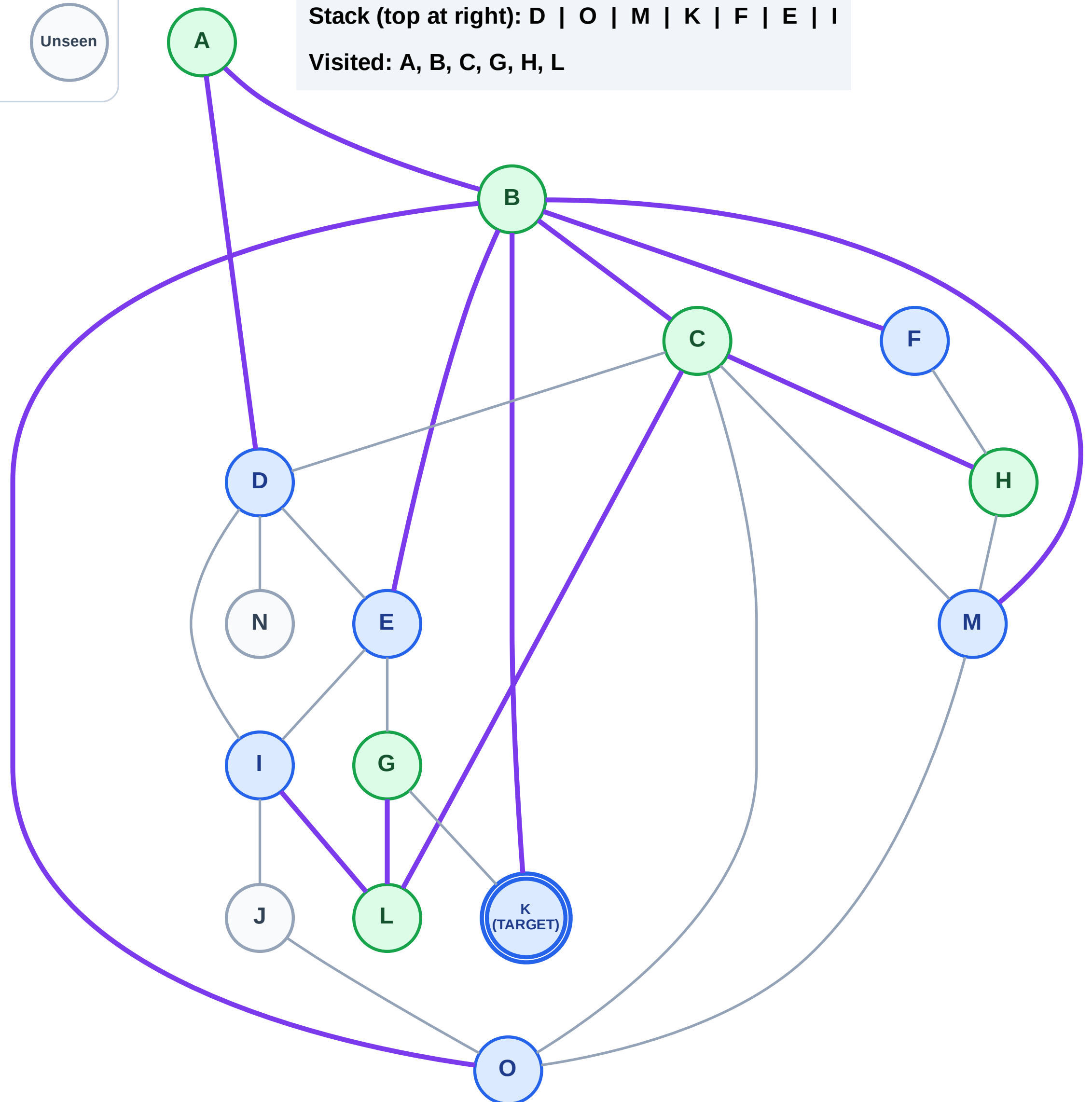


Stack (top at right): D | O | M | K | F | E | I
Visited: A, B, C, H, L



Step 13: No new neighbors from G

Step 13: No new neighbors from G



DFS Traversal

Step 14: Process node I

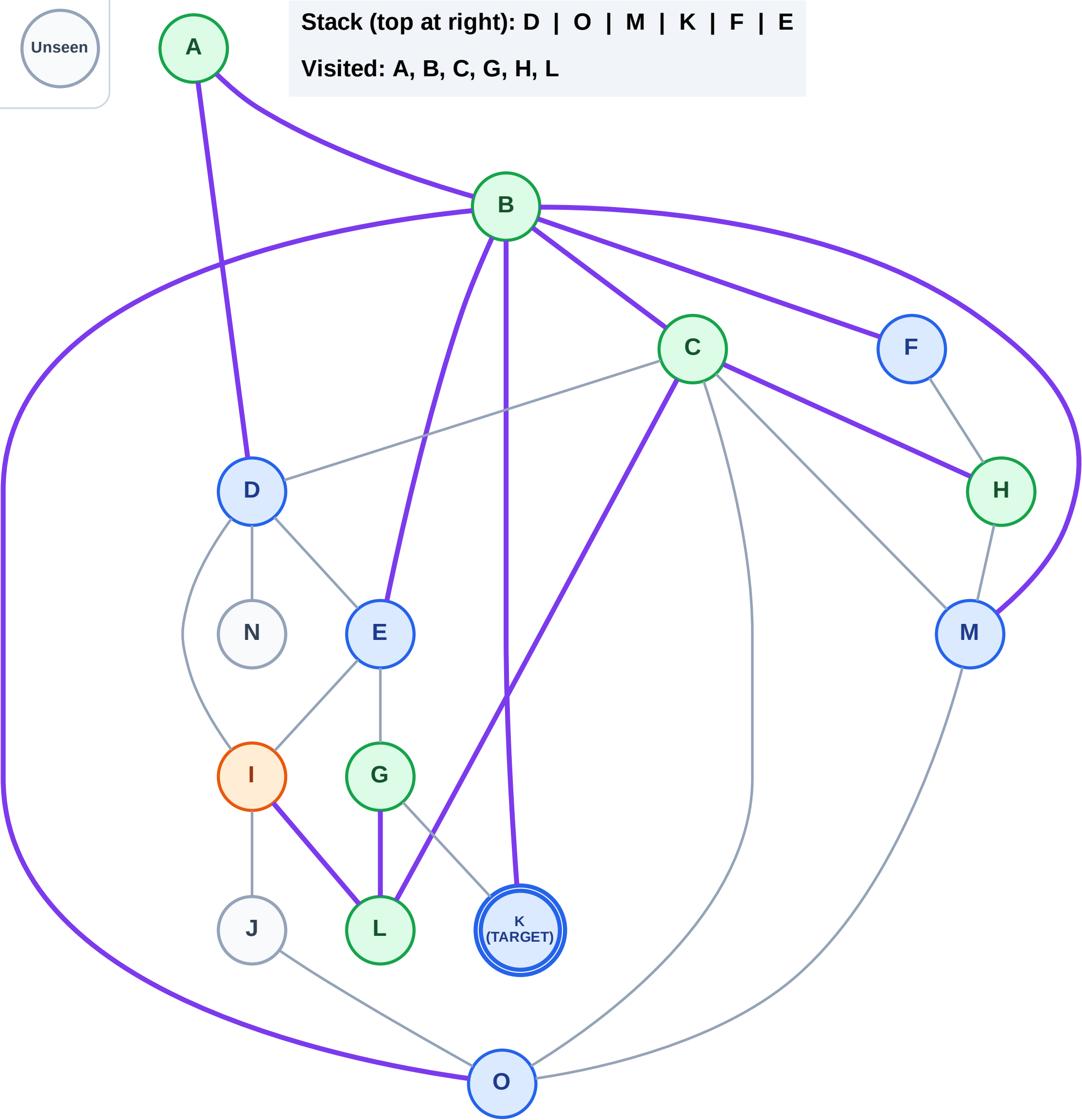
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 15: Discover J from I

Legend

Complete

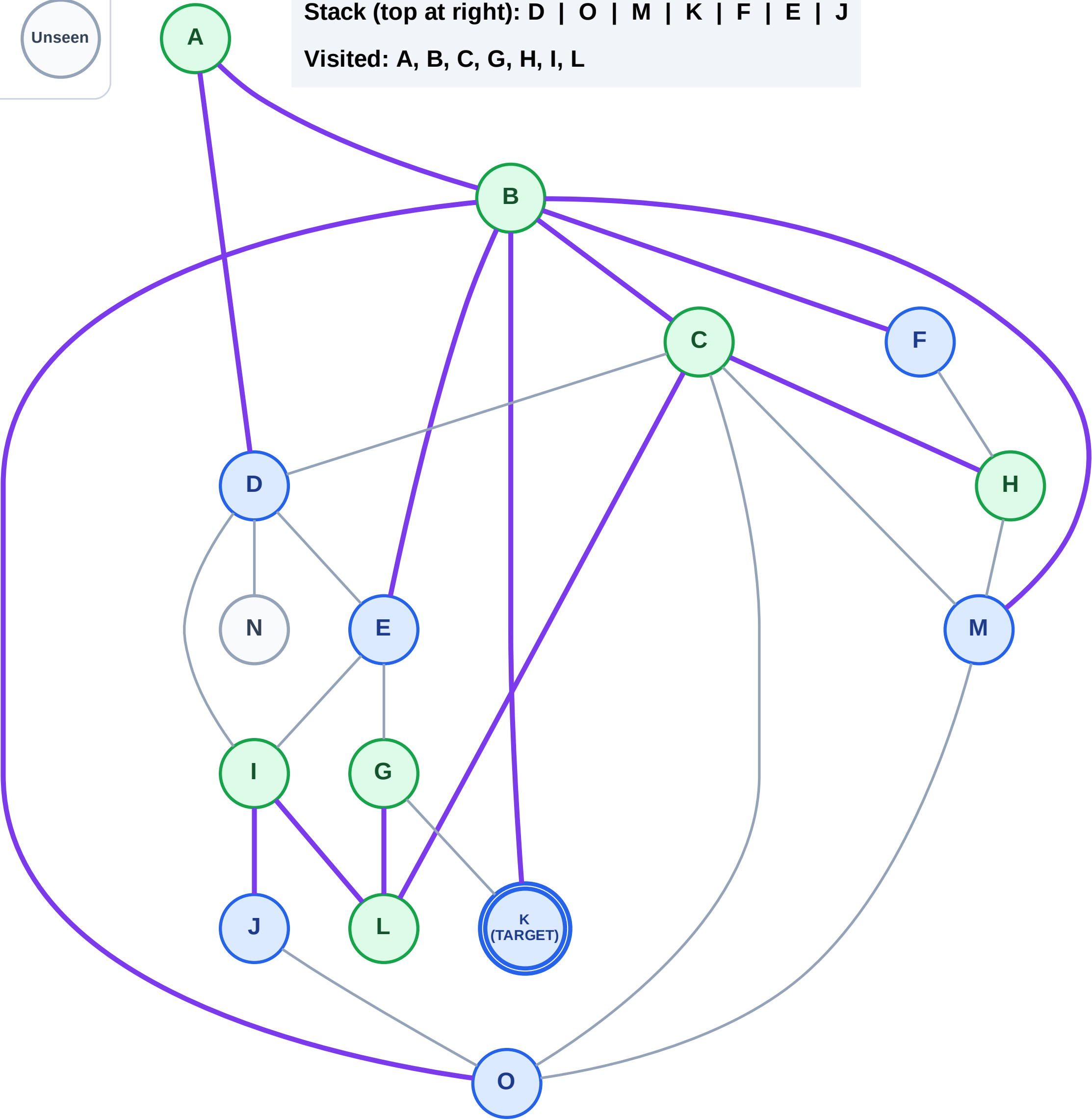
Processing

Discovered

Unseen

Stack (top at right): D | O | M | K | F | E | J

Visited: A, B, C, G, H, I, L



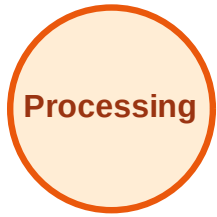
DFS Traversal

Step 16: Process node J

Legend



Complete



Processing



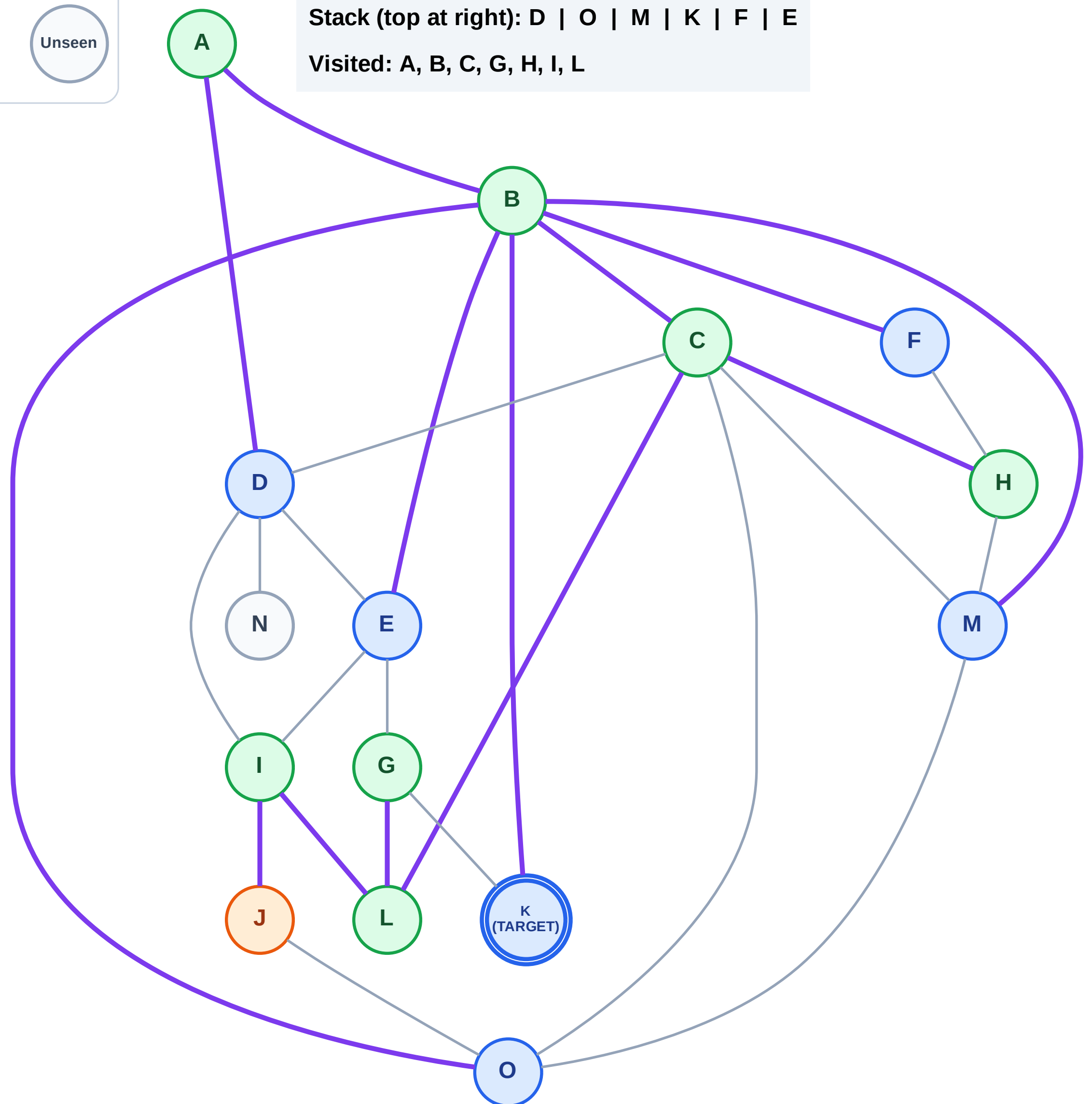
Discovered



Unseen

Stack (top at right): D | O | M | K | F | E

Visited: A, B, C, G, H, I, L



DFS Traversal

Step 17: No new neighbors from J

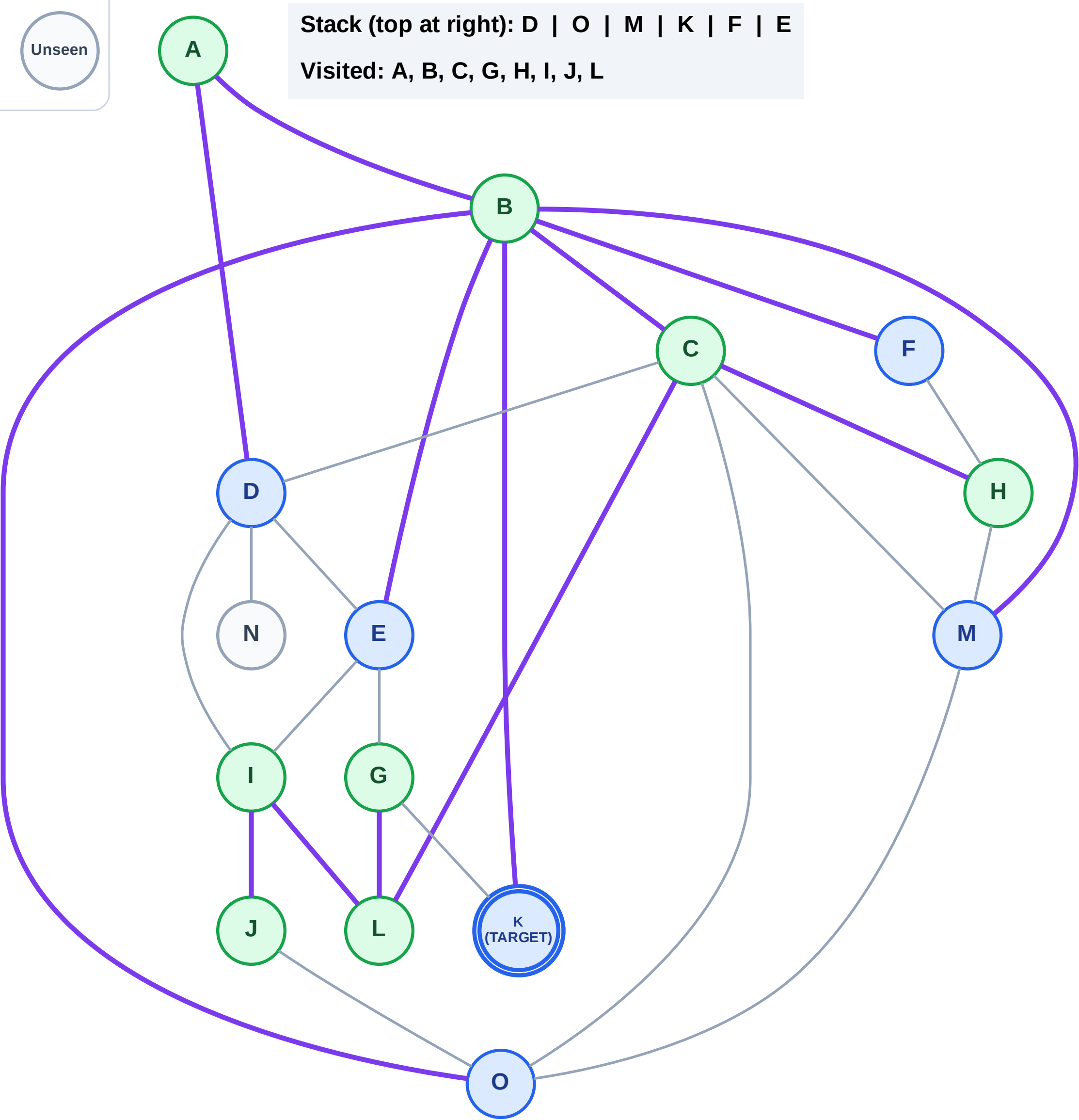
Legend

Complete

Processing

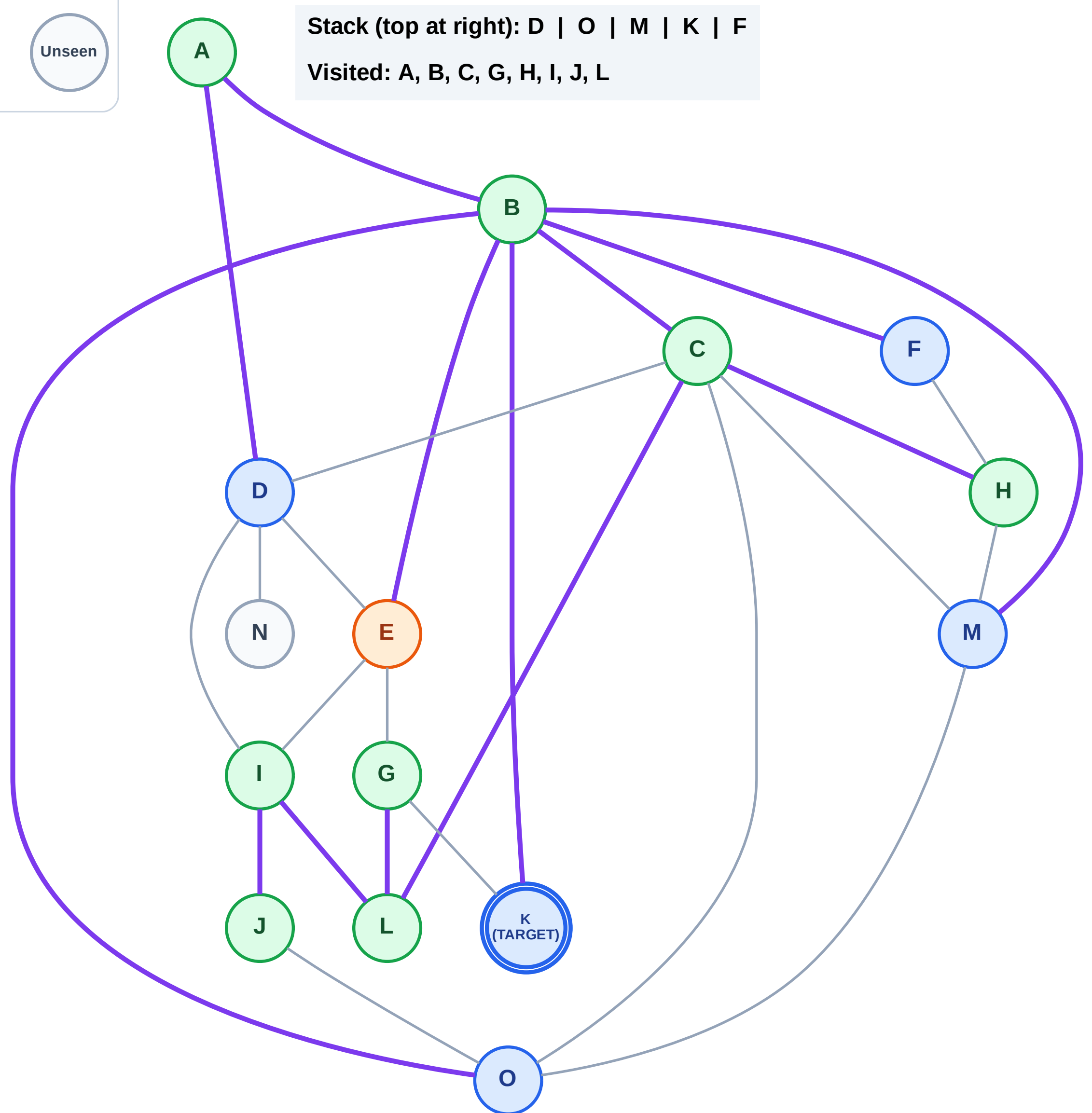
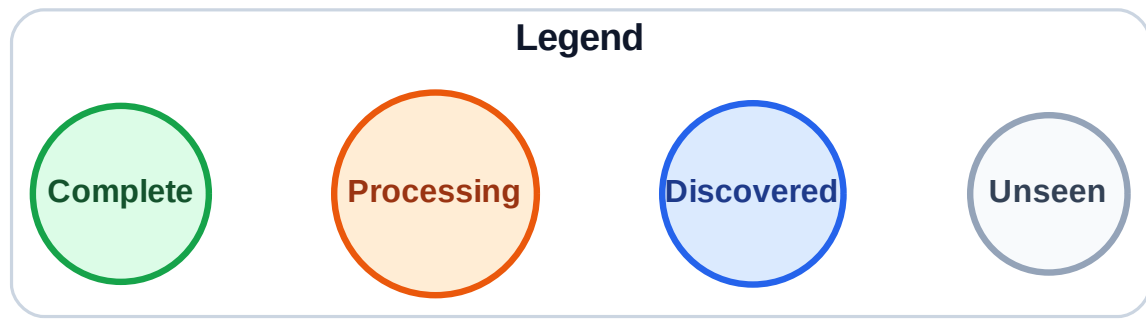
Discovered

Unseen



DFS Traversal

Step 18: Process node E



DFS Traversal

Step 19: No new neighbors from E

Legend

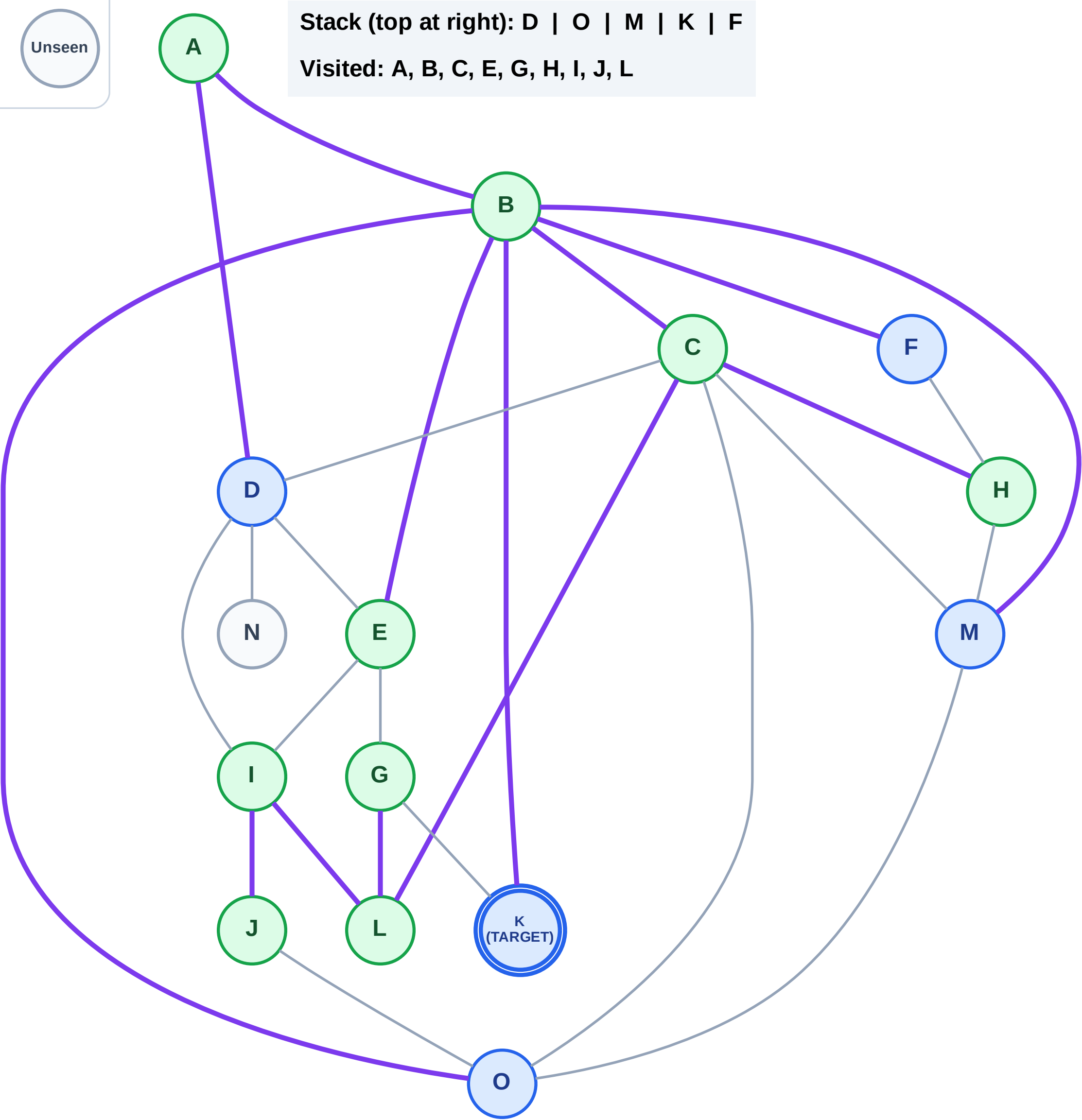
Complete

Processing

Discovered

Unseen

Stack (top at right): D | O | M | K | F
Visited: A, B, C, E, G, H, I, J, L



DFS Traversal

Step 20: Process node F

Legend



Complete



Processing



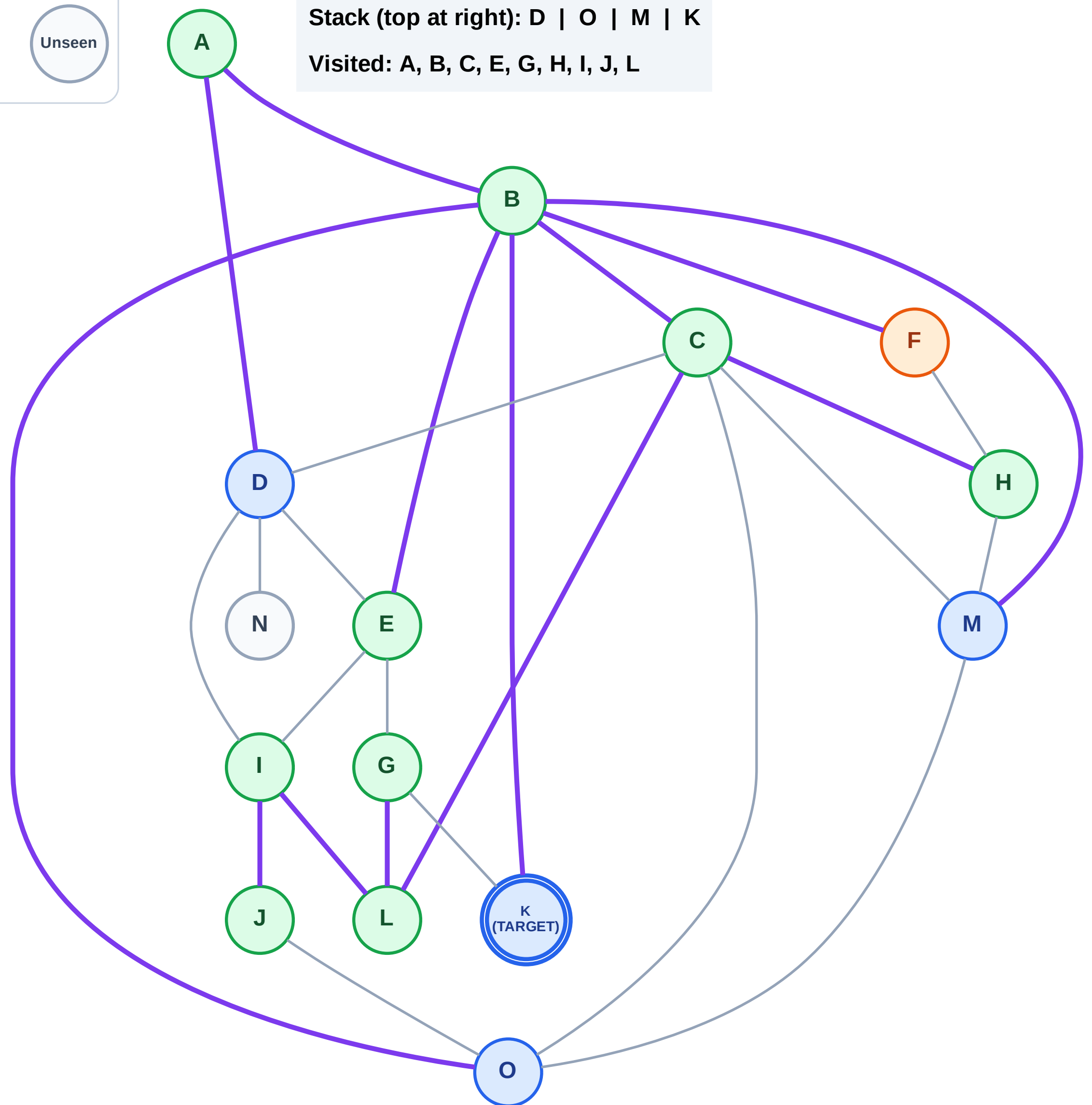
Discovered



Unseen

Stack (top at right): D | O | M | K

Visited: A, B, C, E, G, H, I, J, L



DFS Traversal

Step 21: No new neighbors from F

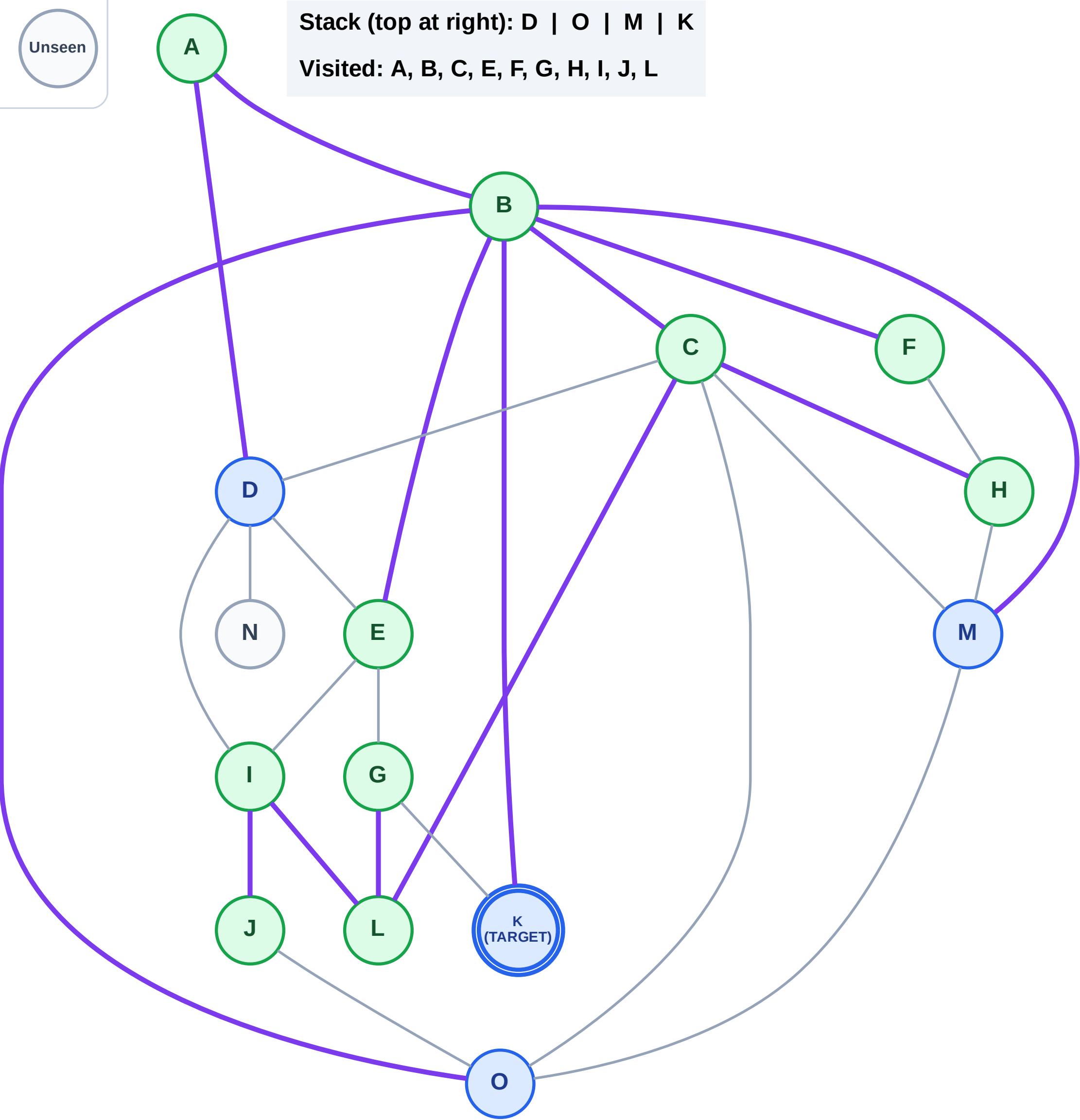
Legend

Complete

Processing

Discovered

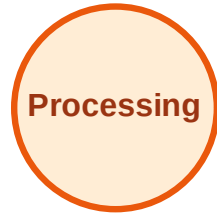
Unseen



DFS Traversal

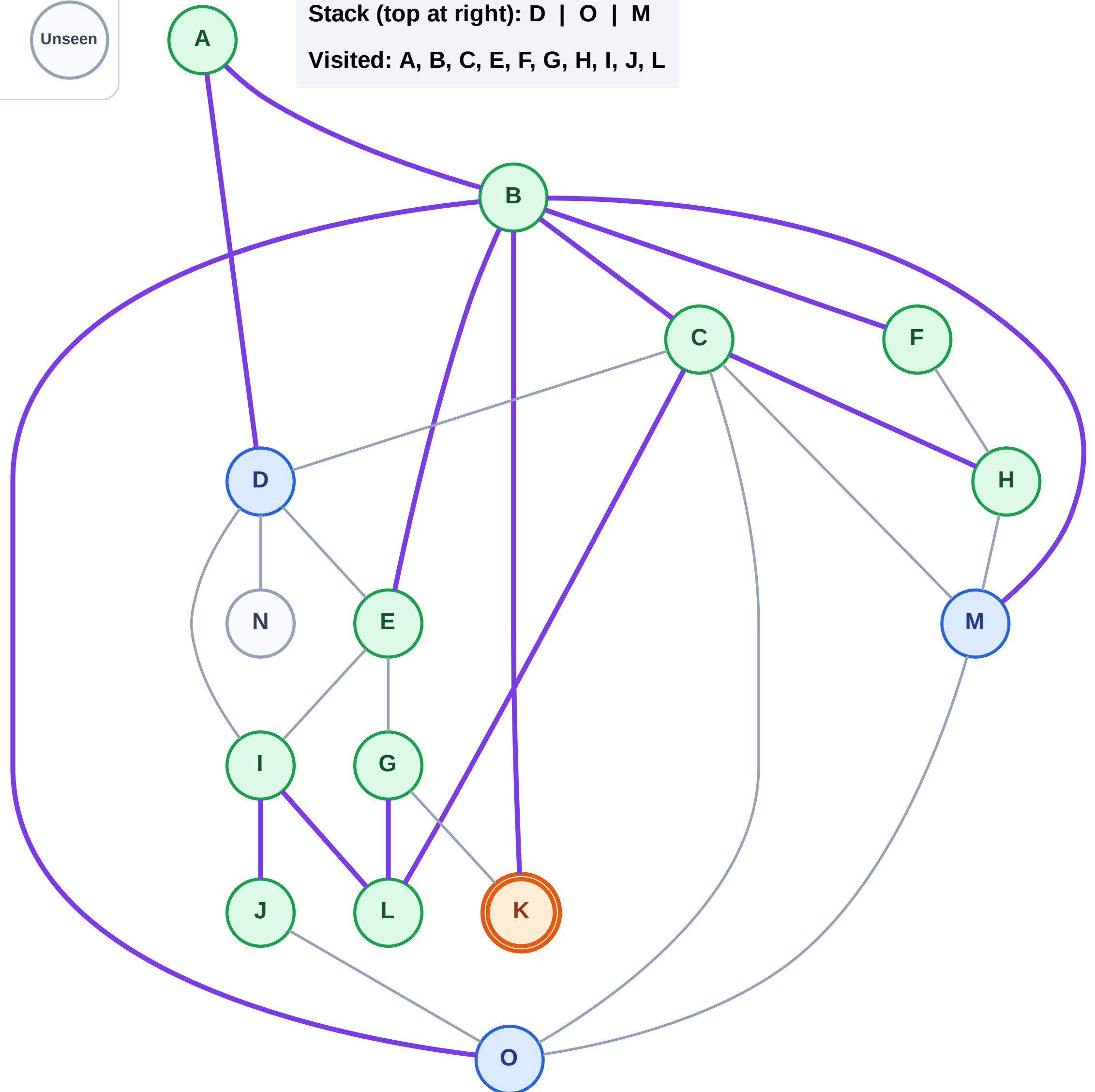
Step 22: Process node K

Legend



Stack (top at right): D | O | M

Visited: A, B, C, E, F, G, H, I, J, L



DFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

